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Master Thesis

Tendon-Driven Mechanisms for Adaptive
Robotic Grasping and Handheld Surgical
Instruments

Author:
Yu WANG

Supervisor:
Prof. Peng LU
Co-Supervisor:
Prof. James LAM

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Tendon-Driven Mechanisms for Adaptive Robotic Grasping and Handheld Surgical Instruments

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Yu WANG

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This thesis studies tendon-driven mechanisms for adaptive robotic manipulation in constrained environments. It concentrates on four mechanism-level challenges: predictable tension transmission, dexterous motion with low distal mass, compliant distributed contact, and reliable operation under tight geometric and ergonomic constraints.

Two prototype embodiments validate the proposed design framework. First, Lasso Gripper is a string-loop end-effector that forms an adaptive capture region via controlled launch-and-retract of a tensile element. We present the mechanical design, control strategy, and demonstration experiments that show the concept’s feasibility and practical advantages over conventional antipodal grippers: an extended capture envelope, distributed contact that reduces localized stress on soft objects, and robustness to positional uncertainty. The experiments reported are demonstration-focused and qualitative; they illustrate mechanism behaviors and common failure modes rather than providing exhaustive statistical metrics.

Second, a compact handheld motorized laparoscopic instrument translates the same tendon-driven principles into a clinically constrained form factor. A benchtop prototype integrating proximal actuation, differential tendon routing, tension-preserving reels, and closed-loop control is described and validated in interaction and interface tests. Clinical evaluation are beyond this work’s scope and are left for future study.

Together, these contributions establish mechanism-level design rules for tendon-driven systems and demonstrate their transferability between general robotic grasping and minimally invasive surgical tools.

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by

Yu WANG

B.Eng. Auburn University, AL, U.S.A

M.S. The University of Hong Kong, Hong Kong

A Thesis Submitted in Fulfilment
of the Requirements for the Degree of
Master of Philosophy

at

The University of Hong Kong

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Declaration

I, Yu WANG, declare that this thesis titled, “Tendon-Driven Mechanisms for Adaptive Robotic Grasping and Handheld Surgical Instruments”, which is submitted in fulfillment of the requirements for the Degree of Master of Philosophy, represents my own work except where due acknowledgement have been made. I further declared that it has not been previously included in a thesis, dissertation, or report submitted to this University or to any other institution for a degree, diploma or other qualifications.

Signed: Yu WANG

Date: May 8, 2026

For Mama and Papa

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List of Publications

Journals:

- [1] R Peng, Y Wang, M Lu, P Lu, “A dexterous and compliant aerial continuum manipulator for cluttered and constrained environments,” Nature Communications, January 2024.
- [2] R Peng, Y Wang, P Lu, “A Tendon-Driven Continuum Manipulator With Robust Shape Estimation by Multiple IMUs,” IEEE Robotics and Automation Letters, 2024, 9(4), 3084 - 3091.
- [3] W Guan, P Chen, H Zhao, Y Wang, P Lu, “EVI-SAM: Robust, Real-time, Tightly-coupled Event-Visual-Inertial State Estimation and 3D Dense Mapping,” Advanced Intelligent Systems, 2024, 1-24.2024, 1-24..

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- [1] Q. Qiao*, Y Wang*, X. Fan, and P. Lu, “Lasso Gripper: A String Shooting-Retracting Mechanism for Shape-Adaptive Grasping,” in Proceedings of the IEEE International Conference on Robotics and Biomimetics (ROBIO), 2025.

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Chapter 1

Introduction

Chapter summary

This chapter motivates the study by describing scenarios where rigid-link manipulators are limited (narrow workspaces, fragile or irregular objects, and applications that require low distal inertia). It states the central research question: how can tendon-driven mechanisms be designed to provide adaptive, dexterous, and reliable manipulation under these constraints.

The chapter also outlines the thesis structure and main contributions: (1) a mechanism-level study of tension management and proximal actuation; (2) Lasso Gripper concept, prototype, and qualitative demonstrations; (3) extraction of transferable design principles and their translation into a handheld motorized laparoscopic instrument; (4) recommendations and future directions for evaluation and clinical translation.

Robotic manipulation increasingly occurs in constrained environments—narrow workspaces, fragile or irregular objects, and tasks that require low distal inertia. Rigid-link mechanisms often perform poorly under these conditions. Tendon- and cable-driven systems are attractive because they transmit force through lightweight tensile elements, allow motors to be placed proximally, and provide compliance at contact points. However, such systems introduce engineering challenges including slack accumulation, hysteresis, friction losses, coupling between degrees of freedom, and reduced predictability under varying loads.

This thesis adopts a mechanism-centered perspective to address these issues. The central question is how tendon-driven mechanisms can be designed to deliver adaptive, dexterous, and reliable manipulation while meeting constraints on size, weight, and compliance. To answer this, a transferable tendon-driven design framework is developed and validated with two embodiments: Lasso Gripper (an adaptive loop-based end-effector) and a handheld motorized laparoscopic instrument.

The first embodiment, termed Lasso Gripper, studies how a flexible tensile element can function not only as a transmission component but also as the primary structure for grasp formation. Inspired by traditional capture tools such as the cowboy lasso and

the Mongolian Uurga, shown in Fig.1.1 and Fig.1.2, the proposed mechanism launches and retracts a controllable string loop to create a large adaptive capture region around the target. This approach differs fundamentally from conventional antipodal grippers, which apply concentrated contact forces over limited areas and often struggle with large, soft, or irregularly shaped objects. By contrast, the string loop enables distributed contact, greater tolerance to positional uncertainty, and a favorable balance between grasp adaptivity and force transmission.



Figure 1.1: Traditional cowboy lasso.

Within this thesis, Lasso Gripper serves as the first mechanism instantiation and as the main source of transferable design insights. Its development makes it possible to study several mechanism-level questions: how tension can be maintained during reversible actuation, how launch-and-retraction dynamics affect capture range, how a flexible loop can improve robustness to object uncertainty, and how remote actuation can be coordinated to preserve low distal complexity. These insights are treated not as properties of a single prototype, but as candidate design rules for tendon-driven systems more broadly.

The second embodiment translates these mechanism insights into a much more constrained application domain. In laparoscopic surgery, surgeons manipulate long and slender tools through small incisions, which severely restrict distal dexterity and attenuate direct haptic perception, as illustrated in Fig.1.4. Conventional instruments typically provide only basic grasping and rotation, while more advanced articulated systems must balance dexterity against weight, complexity, and controllability. Tendon-driven actuation is highly relevant in this context because it permits motors to remain proximal while producing multiple distal degrees of freedom. However, the medical setting imposes additional constraints on size, ergonomic balance, repeatability, safety, and motion fidelity.

The handheld laparoscopic instrument presented in this thesis is therefore framed as a second embodiment of the same tendon-driven mechanism framework. Rather than being a separate topic, it functions as a translational validation of the principles derived



Figure 1.2: Traditional Mongolian uurga.

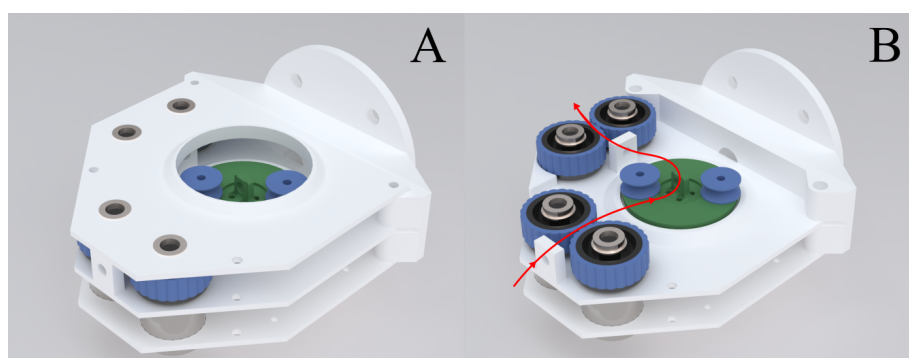


Figure 1.3: Prototype of the proposed Lasso Gripper.

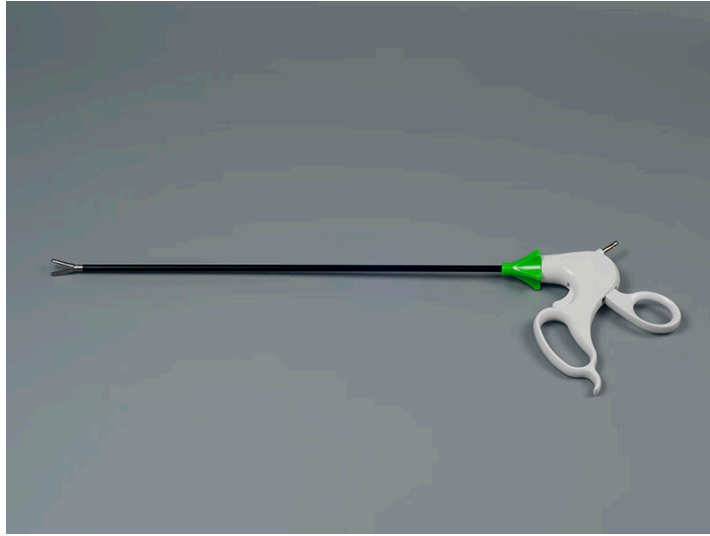


Figure 1.4: Typical single-hole surgical instruments with limited distal dexterity.

from Lasso Gripper. In particular, the instrument tests whether proximal actuation, differential tendon routing, explicit tension management, and sensor-supported closed-loop control can be integrated into a compact handheld system that improves distal dexterity while respecting practical clinical constraints.

The main contributions of this thesis are summarized as follows:

1. A mechanism-level study of tendon-driven robotic systems focused on tension management, proximal actuation, compliant interaction, and low distal inertia.
2. The design, fabrication, and experimental validation of Lasso Gripper as a novel tendon-driven mechanism for adaptive robotic grasping.
3. The extraction of transferable design principles from Lasso Gripper and their translation into a handheld motorized laparoscopic instrument for minimally invasive manipulation.
4. A unified thesis narrative showing how tendon-driven mechanism principles can bridge general robotic grasping and surgical instrumentation.

The remainder of this thesis is organized to follow that mechanism-centered narrative. Chapter 2 reviews related work in laparoscopic end-effectors and adaptive grasping systems, with emphasis on actuation, sensing, and tendon-driven design trade-offs. The subsequent chapters present Lasso Gripper, including its mechanism design, grasping strategy, dynamic behavior, and experimental validation. The thesis then transitions to the handheld surgical instrument, where the previously derived tendon-driven principles are translated into a compact device architecture, cable routing design, sensing strategy, and control framework. The final chapters discuss the broader implications of these results and outline future directions for tendon-driven robotic mechanisms.

Chapter 2

Related Works

Chapter summary

This chapter surveys prior work relevant to adaptive and enveloping grasping as well as tendon-driven laparoscopic end-effectors. It covers loop- and belt-based grippers, sensorized soft grippers, material and microstructure innovations, grasp planning algorithms, actuation technologies, sensing strategies, tip and rod architectures, soft and continuum approaches, and AI-assisted monitoring and control.

The review highlights recurring challenges—such as cable coupling, material creep and stretch, frictional losses, and the trade-offs between flexibility, load capacity, and sensing complexity—and identifies open gaps that motivate this thesis: compact, tension-preserving cable routing; miniature sensing for closed-loop compensation; mechanism-level evaluations linking design choices to performance; and robust sensing and routing strategies for loop-based grippers, which Lasso Gripper seeks to explore.

2.1 Advances in Actuation Technologies for Multi-DOF End-Effectors in Laparoscopic Instruments

Minimally Invasive Surgery (MIS) shifted instrument design toward restoring distal dexterity within severe access constraints. Cable-driven actuation is widely used because it enables multi-degree-of-freedom tips with low distal mass, but it brings challenges such as cable coupling, material creep, and reduced control fidelity. This section summarizes actuation alternatives, sensing strategies for tension and pose estimation, and tip/shaft design patterns that motivate the tendon-driven choices explored in this thesis.

To address the limited dexterity at the instrument tip, a prominent solution involves using cables as a medium to transmit power from the handpiece to the end-effector, thereby enabling additional DOFs, such as roll or yaw rotations. First introduced in the late 1990s, Intuitive Surgical’s da Vinci system exemplifies this approach, employing a cable-driven mechanism to control its articulated tips. The primary advantage of such a system is that cables can transmit power over a distance with a lower weight penalty

compared to rigid linkages or gear trains[2]. However, this benefit is accompanied by inherent challenges. In the initial design of the da Vinci system, cable coupling—where the movement of one cable inadvertently affects another—was a significant issue that could not be overlooked, often leading to kinematic inaccuracies and reduced control fidelity. Therefore, various approaches have been studied to address these problems.

2.1.1 Replacements of Driven Motors

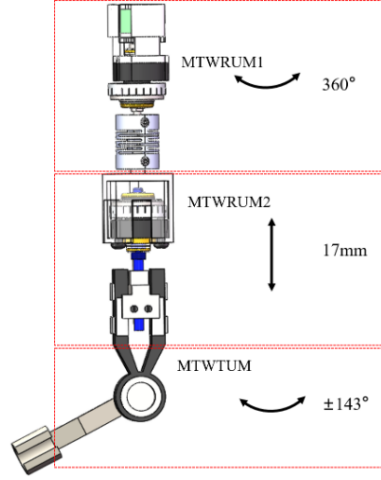


Figure 2.1: Design and illustration of ultrasonic motors[14].

To address the challenges of control fidelity loss and restricted tip articulation in minimally invasive surgical instruments, various driving mechanisms have been extensively investigated. One prominent approach involves the implementation of ultrasonic motors as alternatives to conventional electromagnetic actuators. As illustrated in Fig. 2.1, experimental validation has confirmed that the maximum output torque reaches $1.2 \text{ mN} \cdot \text{m}$ for the rotary configuration and $1.4 \text{ mN} \cdot \text{m}$ for the tilting mechanism. These torque values adequately satisfy the operational requirements of magnetically anchored laparoscopes while offering superior positional precision. Crucially, performance evaluations of the prototype system have substantiated both the feasibility and effectiveness of integrating these ultrasonic motors into laparoscopic applications[12]. This approach presents a novel and practical alternative for achieving enhanced visual platform maneuverability in single-port surgery, addressing fundamental limitations in conventional cable-driven systems through improved torque transmission and positioning accuracy.

Some studies focus on matching the power density and peak torque output of this proposed driving method to those of electromagnetic actuators. By applying a rod-shaped transducer in torsional and bending modes, the rotor is driven in its orthogonal direction with external excitations at specific frequencies. By adjusting the excitation

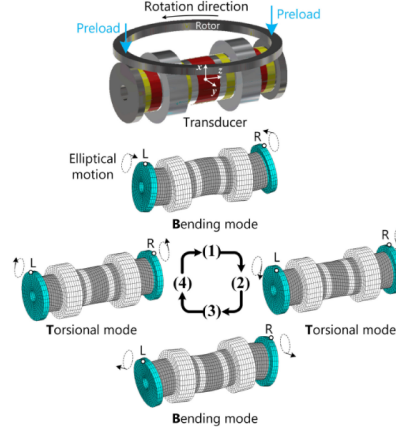


Figure 2.2: Bending and torsion layouts of the transducer[32].

frequencies, the maximum torque recorded in the experiments reached $10.1 \text{ N} \cdot \text{m}$, and the output power reached 38.1 W [32].

2.1.2 Additional Sensors

A fundamental limitation in the original cable-driven system involves the progressive uncertainty of cable length resulting from material creep during prolonged operation. This phenomenon introduces cumulative positioning errors between the input and output ends, significantly compromising control fidelity[18]. To mitigate this inherent uncertainty, the integration of supplemental sensing modalities has been implemented. These sensors enable real-time monitoring of either cable tension, end-effector pose, or both, facilitating a closed-loop control architecture that dynamically compensates for length variations and restores positional accuracy.

Accurate measurement of cable tension during surgical procedures represents a critical safety requirement in cable-driven robotic systems. In traditional laparoscopic surgery, surgeons rely on haptic feedback transmitted through rigid instruments to assess tissue interaction forces. However, in motor-driven systems where actuation forces are generated remotely, this crucial tactile information becomes significantly attenuated. This sensory disconnect necessitates extensive training for surgeons to develop appropriate motor control through the handheld interface, as demonstrated in studies on force sensing in robotic surgery [29].

Force sensing has emerged as a promising solution to restore this lost feedback. The most straightforward approach involves embedding force sensors, such as Fiber Bragg Gratings (FBGs), directly into the surgical instrument. As explored in research on mechatronic design of force sensors[37], this method enables direct measurement of

interaction forces at the end-effector. However, this integration often results in structural redundancy and increased instrument diameter, creating bulky designs that may compromise surgical access and maneuverability in confined operative fields.

To address these limitations, recent research has focused on miniaturized sensing strategies. One innovative approach involves embedding FBGs within precisely machined grooves on inclined cantilevers within the instrument structure[34]. This design exploits the strain measurement capabilities of FBGs to detect minute deformations in the cantilevers caused by cable tension variations. The inclined configuration amplifies these deformations, enabling high-sensitivity force measurement while maintaining a compact form factor compatible with the dimensional constraints of minimally invasive instruments.

2.1.3 Novel Structures of the Tip and Rod Design

In conventional master-slave robotic systems, such as the da Vinci Surgical System and similar handheld robotic instruments, the end-effector typically provides 4 to 5 DOFs. This configuration is designed to replicate the articulation capabilities of the human wrist and fingers, enabling complex maneuvers like pitching, yawing, rolling, and grasping. However, a significant design limitation persists: these multiple DOFs are concentrated exclusively at the distal tip, while the connecting shaft or linkage primarily functions as a passive rotary base.

This recognition of ergonomic limitations in conventional articulated instruments has stimulated significant research into alternative tip architectures in recent years. With advances in the modeling and control of continuum robots, their design principles have been increasingly translated into the medical domain. Unlike traditional tip designs composed of multiple rigid links connected by discrete joints, continuum robots offer a validated scheme for implementing soft or continuously flexible connections—both between the tip and the instrument shaft, and among individual subcomponents of the tip itself. This paradigm shift enables a more distributed deformation along the instrument’s length, effectively increasing the overall dexterity without concentrating all articulation at a single point. The resulting designs, often drawing inspiration from biological structures like tentacles or snakes, promise to mitigate surgeon fatigue by decoupling the end-effector’s pose from the precise position of the surgeon’s hand, thereby offering a more intuitive and ergonomic manipulation platform for complex surgical tasks. Building upon the foundational research in continuum robotics, recent studies have explored innovative structural designs to enhance surgical instrument dexterity and safety. Ranzani et al. (2016) demonstrated a significant advancement by integrating flexible fluidic actuators within a silicone matrix containing multiple pneumatic chambers. This design enabled multidirectional bending and elongation capabilities. The proposed manipulator employed a modular architecture, utilizing granular jamming techniques to achieve dynamic stiffness variation and diverse movement profiles, as illustrated in Fig. 2.3. Further extending this paradigm, researchers have investigated the transition from rigid

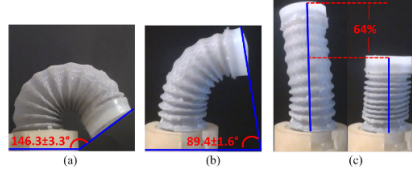


Figure 2.3: Flexible fluidic actuators with granular jamming for variable stiffness [23]

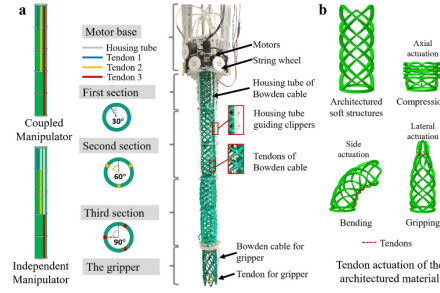


Figure 2.4: The tendon-driven manipulator design with Trimmed Helicoids[9]

linkages to compliant structures to substantially improve operational flexibility. One notable approach involves manipulating the trimmed angle of a helicoid structure to create a variable-stiffness robotic arm[9]. When coupled with a cable-driven actuation system, this design achieves an expanded workspace while significantly minimizing the risk of iatrogenic tissue damage during surgical procedures, as shown in Fig. 2.4. These developments in soft robotics and variable-stiffness mechanisms represent promising directions for next-generation surgical instruments that balance precision with patient safety.

2.1.4 AI-Assisted Monitoring and Control

As analyzed in previous sections, contemporary research in medical instrumentation predominantly advances along three interconnected fronts: innovative tip design, integrated sensory feedback systems, and intelligent control algorithms. The rapid evolution of Deep Learning and Reinforcement Learning methodologies has now catalyzed a fourth research direction: the integration of data-driven approaches into clinical practice.

A particularly promising development lies in repurposing the existing endoscopic camera—initially intended solely for surgical visualization—as an external verification system for instrument navigation. Specifically, Kong et al. (2023) demonstrated that through a specially designed architecture, both the visual stream from the camera and

kinematic data from the actuation motors can be fused to estimate the positional discrepancy between commanded and actual end-effector positions following movement execution[13]. This estimated deviation is subsequently converted into compensation signals that drive automatic self-correction maneuvers, effectively creating a vision-enhanced closed-loop control system that continuously calibrates the instrument without requiring additional hardware beyond the standard surgical setup.

This methodology represents a significant paradigm shift from traditional model-dependent control strategies, leveraging the wealth of available surgical video data to address inherent system limitations such as cable stretch, hysteresis, and kinematic modeling inaccuracies. The approach demonstrates how computer vision and deep learning techniques can transform existing surgical infrastructure into a self-calibrating system, potentially enhancing both positioning accuracy and long-term reliability in cable-driven surgical instruments.

2.2 Development of Adaptive Grasping and Its Applications

2.2.1 The Necessity of Enveloping and Encircling Grasping

Enveloping and encircling grasping provide enhanced stability and shape adaptability for a wide range of objects. This subsection reviews recent structural and control approaches that achieve enveloping grasps and highlights the remaining limitations that motivate loop-based and tendon-driven solutions.

One notable study introduces a dual-mode soft robotic gripper designed with integrated annular structures and a contraction actuator. This design significantly improves shape tolerance and gripping force by altering the contact method between the gripper and the object, adjusting the stiffness and load capacity through controlled material properties [10].

Another novel technique employs a soft gripper that leverages a belt loop with an actuated adhesion mechanism. This mechanism provides bidirectional force, both radial and tangential, which enhances the effectiveness and stability of the grasping action. The minimal interaction force strategy ensures gentle handling, making it suitable for fragile objects [4].

Additional studies introduce an affordable, underactuated compliant gripper capable of adapting to objects of diverse sizes and shapes. The design features a flexure motion transmission mechanism and multifunctional jaws, allowing for a more conforming grasp. This interaction between the gripper and the target object increases both gripping force and stability [26].

Additionally, a study on a tendon-driven robotic gripper highlights the importance of stable enveloping grasps. By optimizing tendon routes and stiffness, this gripper can perform multiple grasping modes, including parallel, enveloping, and fingertip grasps. This versatility ensures the gripper's flexibility and stability across various applications [5].

Enveloping and encircling grasping techniques are crucial in robotic grasping technology. They enhance shape adaptability, gripping force, and stability, providing robust support for robots in complex and dynamic environments.

2.2.2 Grasping System with Integrated Sensors

The integration of sensors with soft materials such as loops significantly enhances the efficiency and adaptability of robotic grasping systems. Various studies have demonstrated the potential benefits of combining these elements to improve performance in different contexts.

A notable example is a modular flying gripper that utilizes a loop gripper mechanism to control its position and pose flexibly. This system, composed of quadrotor-based modules, showcases the versatility of encircling grasping techniques. The decentralized control approach enables the flying gripper to modify its aperture angle, thereby facilitating efficient object grasping and transportation [8].

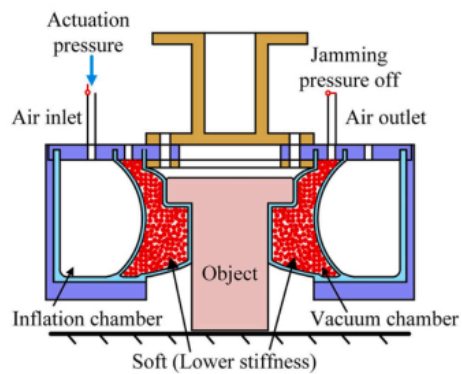


Figure 2.5: Pressurized actuation chamber and adaptive grasping [10]

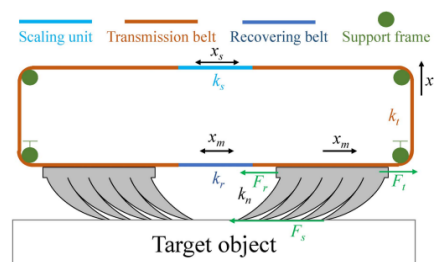


Figure 2.6: Belt-loop actuated adhesion design [4]

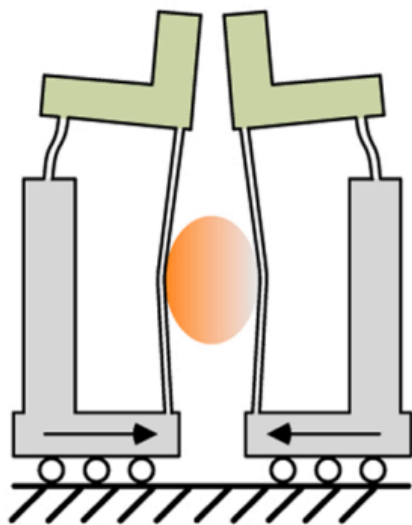


Figure 2.7: Enveloping grasps of the multifunctional jaw [26]

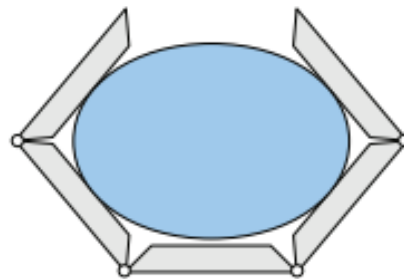


Figure 2.8: Examples of enveloping grasps by the Velo gripper [5]

In another study, a soft robotic gripper equipped with a liquid-metal sensing skin provides a robust feedback mechanism. This gripper, integrated with proximity, pressure, and orientation sensors, operates using shape-memory alloy springs to transition between open and closed states. The closed-loop control enabled by these sensors allows the system to detect and respond to external disturbances, enhancing its grasping, release, and transportation capabilities [35].

Further advancements are seen in the development of a continuum body gripper that incorporates flexible latex bladder sensors. By measuring air pressure changes, these sensors provide tactile and proprioceptive feedback, essential for delicate and high-force grasping. This integration allows the gripper to manage a diverse array of objects, ranging from heavy to fragile items, with minimal control input. The sensorized gripper demonstrates high sensitivity and repeatability, making it a cost-effective and durable solution for universal grasping applications [11].

Additionally, a visual-tactile fusion framework has been proposed to address the challenges of grasping transparent objects in complex environments. This framework combines visual and tactile data to enhance grasping efficiency. A tactile calibration technique combined with a visual-tactile fusion classification method greatly enhances the success rate and accuracy of grasping transparent objects. The experimental results validate the effectiveness of integrating tactile sensors within the gripper system, particularly for objects that are irregular or visually undetectable [16].

The combination of sensors with soft materials such as loops in robotic grippers offers significant improvements in grasping systems. These integrations provide enhanced feedback, adaptability, and control, rendering them invaluable for a wide array of applications, from delicate object handling to complex environmental interactions.



Figure 2.9: The Flying Gripper holding a coffee cup [8]

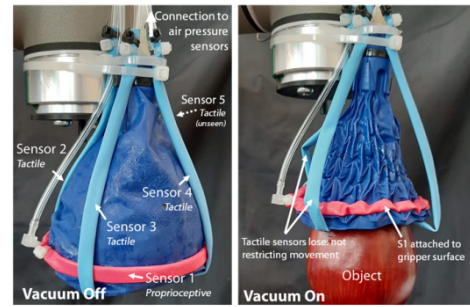


Figure 2.10: Illustration of the tactile sensor and pressure sensors integrated into the loop gripper [11]

2.2.3 Designing for the Physical Properties of Soft Grippers

Through advancements in material technology and microstructural design, significant progress has been made in altering the physical properties of flexible materials like

ropes. These innovations have paved the way for the development of more efficient and adaptable soft robotic grippers.

Wang et al. (2021) introduced a groundbreaking soft gripper using a self-sensing super-coiled polymer (SCP) actuator. This actuator is crafted from multi-strand sewing thread that undergoes a self-coiling process, enabling it to untwist when heated. The addition of a metal wire wrapped around the SCP enhances temperature measurement precision, forming a closed-loop control system for accurate gripper manipulation. This approach offers a foundation for customizing the physical properties of rope-like materials for actuator applications [30].

Sun et al. (2020) described a soft gripper with adjustable stiffness, inspired by the behavior of pangolin scales. The gripper is composed of two layers: a variable stiffness layer and a pneumatic-driven layer. The variable stiffness layer emulates pangolin scales, remaining flexible during normal activities but becoming rigid when threatened. The toothed pneumatic actuator increases the gripper's stiffness, enabling it to grasp a wide range of objects effectively. The autonomous control system of this gripper collects feedback signals to adjust the stiffness autonomously, making it suitable for unstructured environments [25].

Zhang et al. (2017) tackled the design problem of soft robots using a topology optimization framework. A pneumatically actuated soft gripper was developed, with three fingers each designed as a continuum compliant mechanism to achieve maximum bending deformation. This actuator was fabricated through 3D printing, and experimental results demonstrated its ability to grasp objects effectively. This optimization approach provides valuable insights for further exploration of the microstructure of lasso grippers [38].

Li et al. proposed a soft gripper inspired by the inchworm's true leg. This design includes two rows of sharp hooks on a soft palm, with connections made via Shape Memory Alloy (SMA) springs. The design allows the gripper to adhere to rough surfaces and grasp objects with varying shapes and dimensions. The combination of elastic muscle-like structures and sharp hooks enhances the load-carrying capacity of the gripper, demonstrating the potential of microstructural design in soft robotics [15].

Wang et al. (2017) explored the use of Shape Memory Alloy (SMA) in soft grippers, addressing the inherent softness and low actuation force of traditional systems. The study described an SMA-based soft gripper with variable stiffness, composed of three fingers, each with two hinges. The hinges can achieve a significant change in stiffness, allowing the gripper to adapt to various shapes and weights. This material modification approach shows promise for effective grasping applications [31].

The integration of advanced material technologies and microstructural designs has significantly enhanced the capabilities of soft robotic grippers. These innovations provide a robust foundation for developing adaptable and efficient grasping systems.

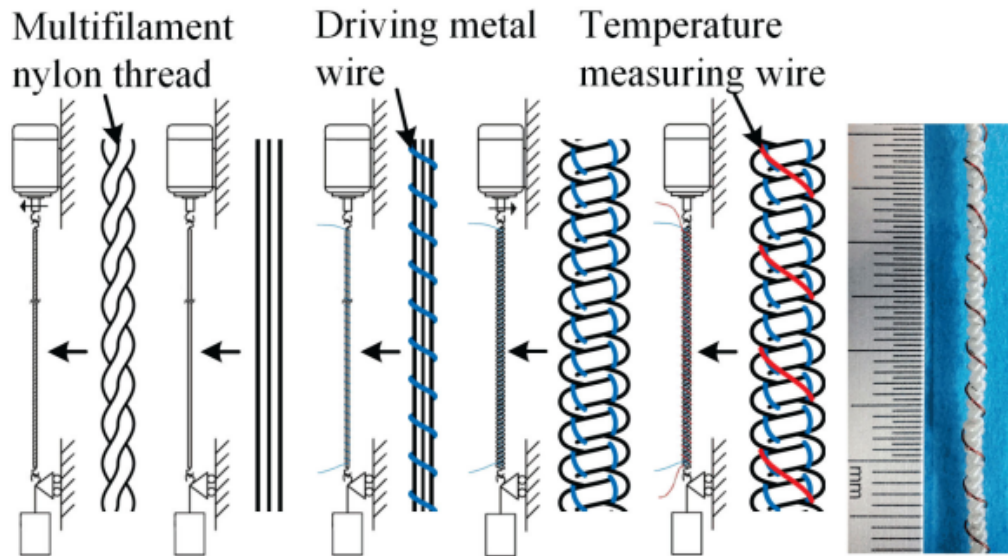


Figure 2.11: Manufacturing process of actuator with multifilament nylon thread [30]

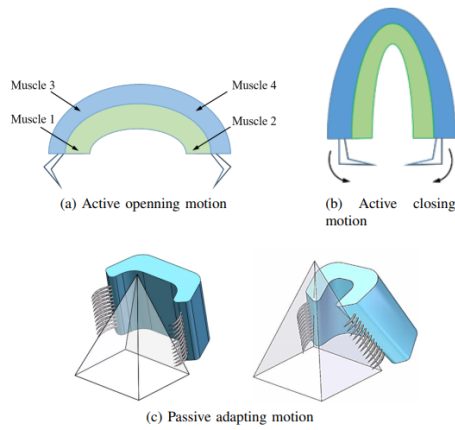


Figure 2.12: Bio-mechanism of inch-worm's true leg [15]

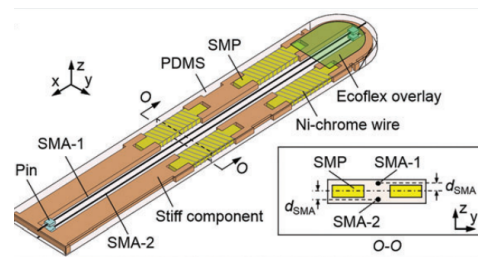


Figure 2.13: Shape Memory Alloy-Based Soft Gripper [31]

2.2.4 Advancement of Grasp Planning Algorithms for Lasso Grippers

The design of grasping algorithms has made significant strides in enhancing the adaptability and efficiency of robotic grippers, particularly those employing rope-like structures such as lasso grippers. These algorithms leverage unique approaches to optimize grasp planning and ensure stable and robust manipulation.

Li et al. (2015) proposed an innovative grasp planning method that addresses the computational challenge of determining stable force-closure grasp configurations for multi-fingered hands. A low-level shape exploration technique is employed in this method by wrapping multiple cords around an object to quickly identify promising grasping regions. Using a variant of the close-until-contact procedure, contact points are determined by the algorithm, which then utilizes finger kinematics and contact information to filter out unstable grasps. This approach not only enhances the natural appearance of grasps but also improves efficiency, as it bypasses extensive preprocessing steps like segmentation and medial axis extraction. The adaptability of this method suggests that the structural concept of wrapping cords around objects can be effectively applied to lasso grippers, potentially offering superior grasping performance [17].

Fang et al. (2023) introduced the AnyGrasp algorithm, designed to enhance robust and efficient grasp perception across both spatial and temporal domains. This algorithm utilizes a dense supervision strategy, integrating real perception data with analytic labels to create a comprehensive training dataset. By incorporating object center-of-mass awareness into the learning process, AnyGrasp improves grasp stability. Additionally, it employs grasp correspondence across observations, enabling dynamic tracking of grasp poses, which ensures effectiveness even in noisy environments. The success of AnyGrasp in achieving a high success rate in bin clearing tasks underscores its potential applicability to various robotic grippers, including those with lasso structures [7].

These advancements in grasping algorithm design underscore the importance of innovative computational methods in enhancing the functionality of robotic grippers. The adaptive planning optimization demonstrated by Li et al. (2015) highlights the potential of rope-like structures in improving grasp performance. As the field continues to evolve, integrating these approaches with lasso grippers could lead to more versatile and effective robotic grasping systems.

2.2.5 Application of Lasso Grippers in Real-World Scenarios

The application of lasso grippers in various real-world scenarios demonstrates their versatility and effectiveness in handling a variety of objects with minimal control effort and high reliability. This section reviews recent advancements and practical implementations of lasso grippers across different fields.

Manes et al. (2024) introduced a soft cable loop-based gripper designed for robotic automation in chemistry laboratories. This gripper can reliably grasp cylindrical and prismatic objects of various sizes with minimal clearance. The design uses the compliance of a cable to completely envelop target objects, with support provided by a rigid finger

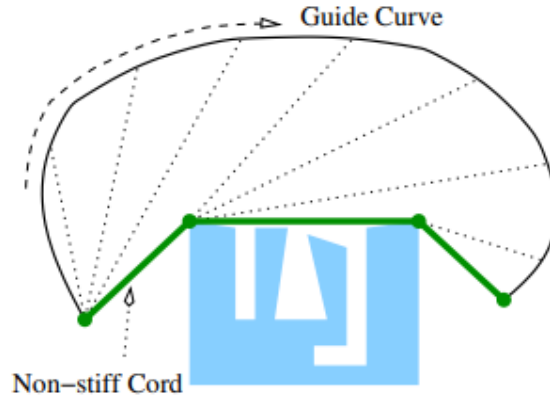


Figure 2.14: Geometric construction of a non-stiff cord [17]

once the grip is secured. This mechanism requires minimal control effort, making it highly practical for repetitive tasks in automated chemistry processes. The gripper demonstrated reliable grasping, with a failure rate of less than 1%, showcasing its efficacy in handling delicate laboratory glassware. However, the use of a self-supporting plastic polymer material limits the loop size, as larger loops would fail to maintain their shape [20].

A cable-driven gripper with perception capabilities for autonomous strawberry picking was developed by Xiong et al. (2018). The gripper utilizes six fingers to create a closed space around a target strawberry, gently cutting the stem without causing damage to the fruit. Equipped with infrared sensors, positional errors can be corrected by the gripper, enhancing robustness and achieving a high success rate in picking isolated strawberries. The integration of internal perception allows the gripper to tolerate positional inaccuracies, reducing the need for slow, high-level closed-loop control. This design highlights the advantages of using cable actuators, such as flexibility and biological friendliness, making it suitable for delicate agricultural applications [33].

Roth et al. (2021) presented the Loop Gripper designed for handling honeycomb materials used in various industries. This gripper uses adjustable threads to provide the necessary flexibility and sensitivity to grip honeycomb panels without damaging their fragile cell walls. Through a series of tests, the optimal gripping parameters for various honeycomb materials were determined, proving the concept's suitability for automated gripping tasks. The Loop Gripper's approach of using a soft, structure-friendly force to grasp objects aligns with the principles of lasso grippers, emphasizing the importance of adaptability and gentle handling [24].

These studies underscore the practical applications of lasso grippers in diverse fields, from chemistry laboratories to agricultural and material handling tasks. The inherent



Figure 2.15: A cable loop gripper prototype evaluated for grasping various types of chemistry glassware [20]

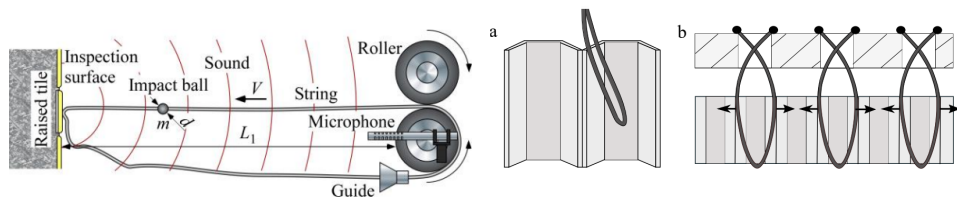


Figure 2.16: String-shoot impact toward a wall [28]

Figure 2.17: String loop gripper for honeycomb materials [24]

adaptability and gentle handling capabilities of lasso grippers make them an effective solution for various automated processes, demonstrating their potential for widespread adoption in real-world scenarios.

2.2.6 The Balance between Flexibility and Strength

Current research in robotic gripping mechanisms often focuses on achieving a balance between flexibility and strength. Pneumatic flexible grippers, for instance, offer excellent shape adaptability but are limited in their working range and load capacity. Yu et al. have noted that soft pneumatic actuators are widely used in soft robots, where their bending angles and motion rules at different pressures are crucial for practical applications [36]. However, despite their flexibility, these actuators struggle with heavier loads and larger workspaces. Moreover, their dependence on precise pressure control can make them susceptible to performance inconsistencies, particularly in environments where maintaining stable pressure is challenging.

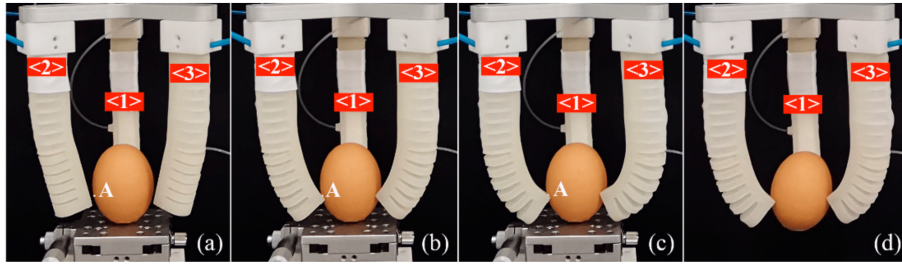


Figure 2.18: Pneumatic actuator grasping process of an egg [36]: (a) The soft actuator moves toward the egg when pneumatic module one is pressurized to 10 kPa, with module two remaining at zero pressure. (b) Contact between the tip of the soft actuator and the egg occurs when the pressure in module one is increased to 20 kPa and module two is set to 14.1 kPa. (c) The egg is securely pinched by the gripper when both module one and module two are pressurized to 20 kPa. (d) The egg is held firmly when the supporting platform is removed, indicating a stable grasp by the gripper.

On the other hand, chameleon-like shooting manipulators offer extended working ranges but suffer from lower success rates and reduced effective working ranges [21]. Such systems often rely heavily on complex control algorithms and precise timing, which can be difficult to manage in real-time applications, leading to potential delays and reduced effectiveness in fast-paced environments.

Fractal grippers, as discussed by Tisdale and Burdick, draw inspiration from an older concept called the Fractal Vise [27]. Their work emphasizes the gripper's ability to conform passively to various objects using a single actuator. Nonetheless, while these mechanisms offer synergistic properties and adaptability, their integration with existing systems and capacity for high-load tasks are still limited. The passive nature of these grippers, while beneficial in certain contexts, can also lead to a lack of control over the

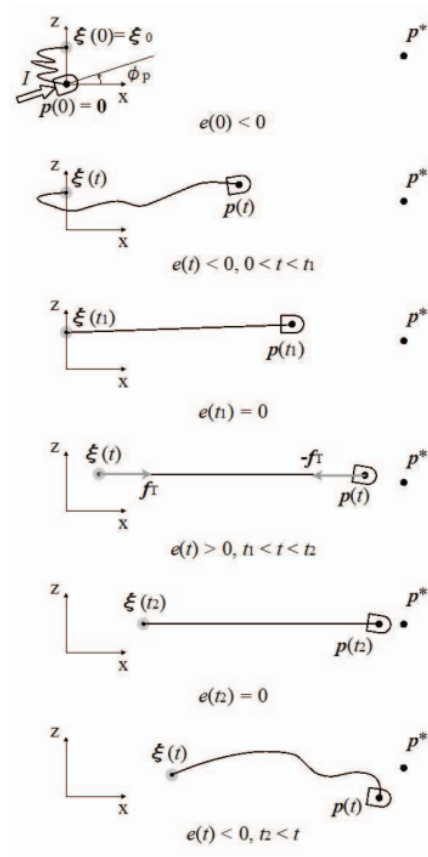


Figure 2.19: Manipulating model of chameleon-like gripper [21].

gripping force, making them unsuitable for tasks requiring precise manipulation of heavy or delicate objects.

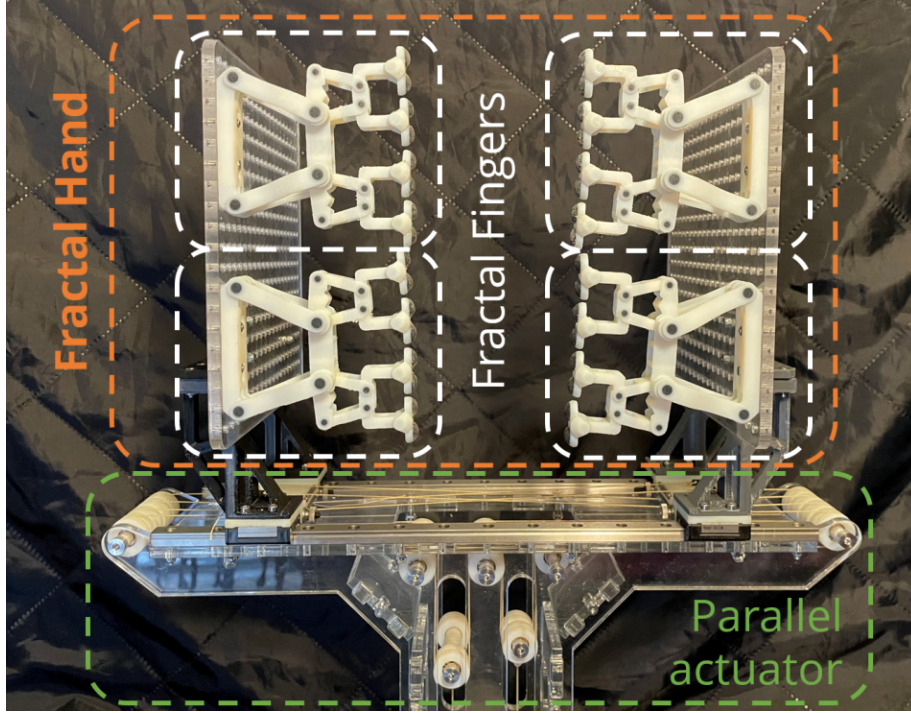


Figure 2.20: Fractal Hand prototype [27].

Lasso Gripper is designed to address these gaps by incorporating a mechanism that can rapidly shoot and retract a string loop, enabling it to capture objects with high speed and precision. Inspired by the traditional lasso used by cowboys and the uurga used by Mongolian herders, this mechanism leverages the structural advantages of a string loop to prevent slippage and ensure a secure grasp. Moreover, Lasso Gripper's design allows for a large initial capture range, significantly increasing the probability of successful grasping, particularly with fast-moving targets.

Brun et al. provide an insightful analysis of the mechanics of the lasso, illustrating how the rope's shape remains stable in the rotating reference frame of the hand, with line tension balanced by centrifugal force and the rope's weight [3]. This fundamental understanding of lasso mechanics informs the design and functionality of Lasso Gripper, ensuring that it can operate effectively under dynamic conditions.

This research will delve into the design and working principles of Lasso Gripper, discussing its overall structure, the specifics of the string shooting-retracting mechanism, and additional design innovations within the gripper. We will also explore the grasping algorithm, which utilizes depth camera scanning and point cloud analysis to accurately locate and secure the target object. Furthermore, we will design experiments comparing Lasso Gripper's performance with other gripping mechanisms, highlighting its advantages in various scenarios.

Chapter 3

Lasso Gripper: Design, Analysis, and Experiments

Chapter summary

This chapter presents Lasso Gripper as the principal mechanism embodiment used to explore tendon-driven design rules. The chapter focuses on the device concept, mechanical architecture, actuation and sensing choices, the grasp planning strategy, and a set of demonstration experiments. The goal is to establish the feasibility of using a controllable tensile loop as a practical end-effector and to identify the mechanism-level advantages and failure modes that inform the broader thesis.

The experimental program reported here is demonstration-focused and qualitative by design: experiments illustrate the gripper’s operating behaviors, its tolerance to object uncertainty, and comparative advantages versus a classical antipodal gripper. Quantitative statistical evaluation (e.g., large-sample success rates, repeatability statistics, fatigue testing) is not included and is recommended as a follow-up study.

3.1 Design and Principle

Lasso Gripper integrates four core subsystems: the launch-and-retract end-effector, a proximal control unit, sensing for tension and loop position, and a host robot for gross positioning. The end-effector implements the friction-wheel propulsion and spool retraction mechanisms; the control unit coordinates motor commands and processes sensor feedback; sensing includes a spool-mounted force sensor and motor/position encoders used for tension regulation; the host robot provides coarse positioning and aiming for capture. This modular decomposition clarifies subsystem responsibilities and guides the experimental evaluation reported below.

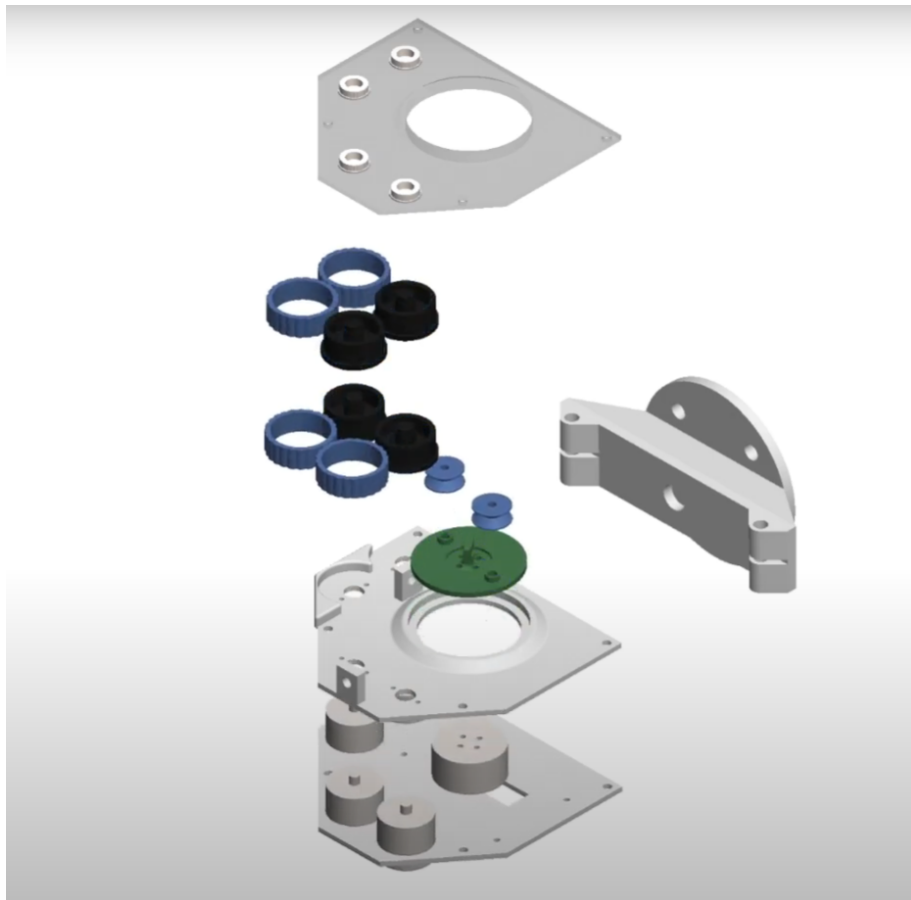


Figure 3.1: The SolidWorks designed hardware of the proposed gripper

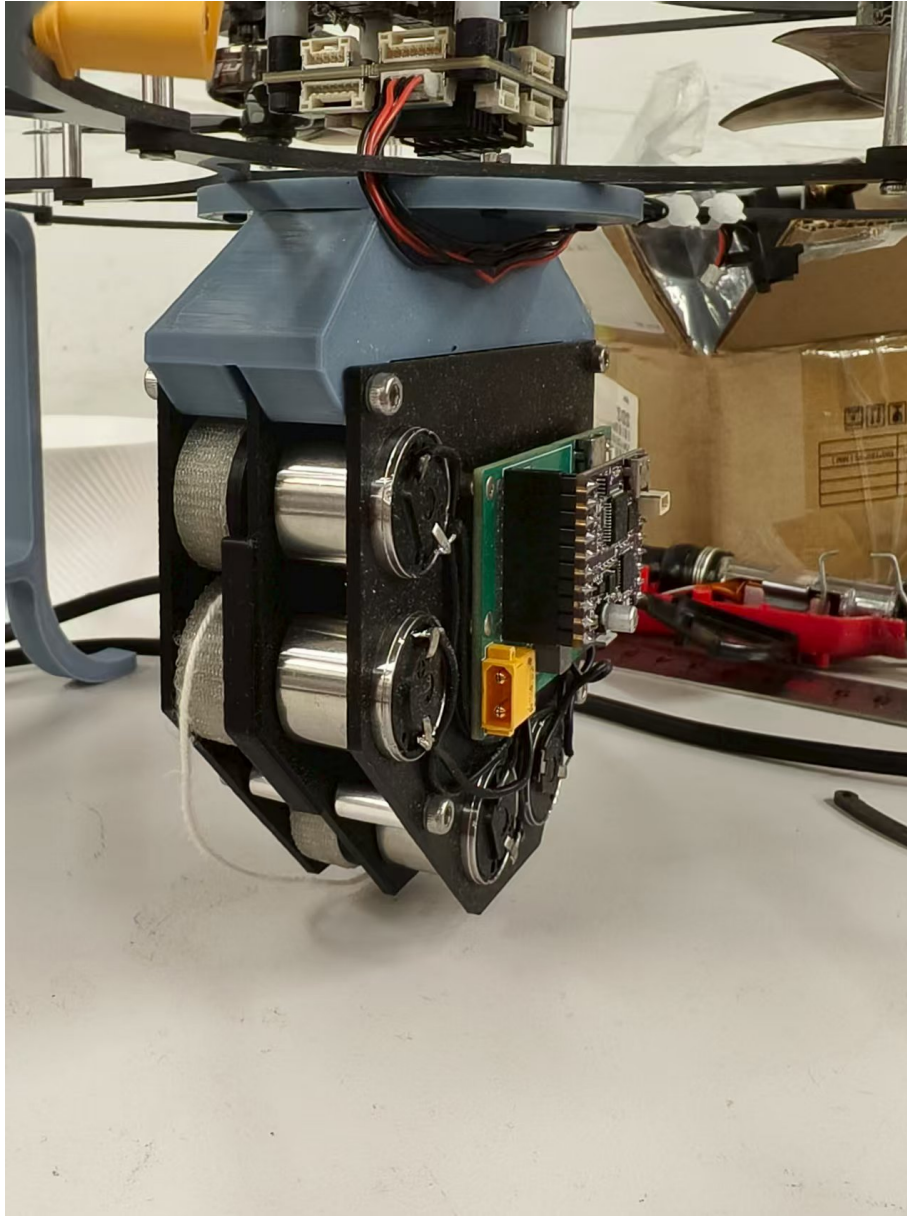


Figure 3.2: The proposed gripper connected with an ESP32 controller, motor electronic speed controller.

3.1.1 String Shooting and Retraction Mechanism

The string shooting and retraction mechanism is a defining feature of Lasso Gripper, enabling it to perform complex gripping tasks with high efficiency. The mechanism comprises multiple components working in unison to achieve precise control over the string's movement.

The propulsion of the string is achieved through two pairs of friction wheels. These wheels are designed with high-friction surfaces to grip the string effectively. The wheels are driven by coreless motors with high speed so that the required rotational speeds for rapid string extrusion can be achieved[22].

During the string shooting phase, both pairs of friction wheels rotate in the same direction, propelling the string forward through a guided track. This track is meticulously designed to ensure the string forms a loop within the desired dimensions and shape. Sensors along the track monitor the string's position and velocity, providing real-time feedback to the controller.

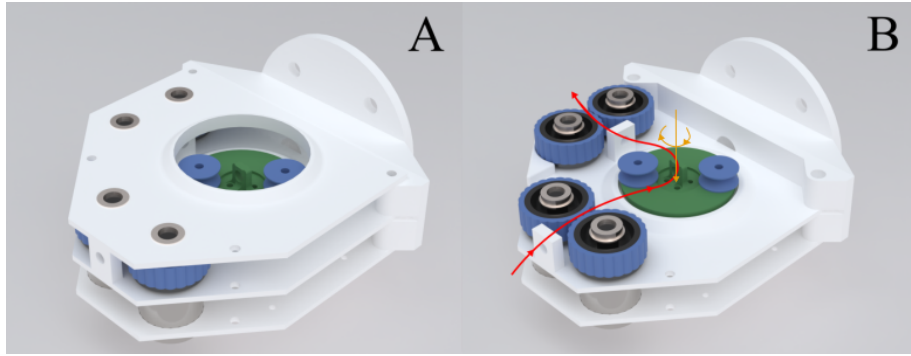


Figure 3.3: The friction wheels and the retraction mechanism driving motors are highlighted for illustration.

When retraction is required, the mechanism undergoes a coordinated sequence of actions. The rear pair of friction wheels maintains its rotational speed, continuing to pull the string inwards. Simultaneously, the front pair of wheels reverses direction, creating a counterforce that pulls the string loop into the space between the two pairs of wheels. This space is precisely designed to accommodate the retracted string without entanglement.

Within this confined space, a spool mechanism is designed. The spool is equipped with a rotational motor that winds the string onto itself, storing it neatly for future use. The tension exerted on the string during retraction is carefully controlled by a force feedback sensor located on the spool's axle. This sensor monitors the tension and adjusts the motor's torque to ensure that the object is gripped securely without applying excessive force that could cause damage.

3.1.2 The ESP32 Main Controller for Lasso Gripper

The ESP32 Main Controller serves as the central hub for Lasso Gripper operations. The ESP32, with integrated Wi-Fi and Bluetooth, provides the computational power and connectivity required by the prototype. Its dual-core processor handles real-time

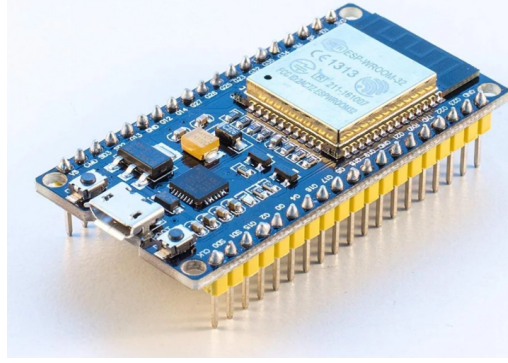


Figure 3.4: The ESP32 controller used in the proposed Lasso Gripper.

sensor processing, execution of the grasping algorithms, and communication with external devices. The controller interfaces with a spool-mounted force sensor, motor encoders, and any auxiliary trackers used during experiments.

GPIO pins drive the retraction modules and friction-wheel controllers, while PWM outputs provide fine speed control for the propulsion motors. Wireless connectivity is used for logging, target-recognition offload, and remote monitoring via a host computer when required.

3.2 Implications for Tendon-Driven Mechanisms and Medical Devices

Work on Lasso Gripper provides several mechanism-level insights that generalize beyond the specific prototype. First, the principle of using a flexible tensile element to create an adaptive capture region suggests new ways to trade capture range for control complexity. Second, differential and proximal actuation patterns enable low distal mass while still producing multi-degree-of-freedom effects at the end-effector; this trade-off is central to designing handheld tools and remote manipulators. Third, performance metrics such as capture probability, peak tensile load, retraction dynamics, and hysteresis in the cable path form a compact evaluation set that is applicable to other tendon-driven designs.

These insights motivated the subsequent handheld surgical instrument presented in this thesis: the instrument intentionally reuses the same tendon-driven design philosophy and control principles, but applies them to the clinical constraints of minimally invasive surgery (low profile, ergonomic balance, safety, and sterilizability). The following chapter

therefore uses Lasso Gripper both as a technical proof-of-concept and as a source of formalized mechanism design rules.

Chapter 4

Gripping Algorithm Design

The design of the gripping algorithm is critical to the functionality and efficiency of Lasso Gripper, ensuring precise and reliable manipulation of various objects. This chapter outlines the algorithm’s design, which leverages practical point-cloud processing, loop-synthesis for caging grasps, and real-time tension feedback to achieve robust performance in demonstrative scenarios.

4.1 Object Recognition and Point Cloud Processing

The gripping algorithm begins with target detection using an Intel RealSense D435i depth camera. The captured point cloud is processed to locate candidate "neck" regions and to synthesize feasible caging loops that avoid reliance on exact surface contacts. The algorithm output is a target loop position and approach plan that the launch-and-retract mechanism can execute.

A concise pseudocode summary:

1. Acquire point cloud P from depth camera.
2. Preprocess P (filtering, downsampling, outlier removal).
3. Identify candidate neck regions using caging-loop detection.
4. Select loop pose L according to capture feasibility and approach constraints.
5. Execute launch sequence and monitor spool tension.
6. Retract and close loop; apply tension-based release criteria.

4.2 Grasping State Machine

The grasping controller is implemented as a simple finite-state machine. States and typical transition conditions are:

- Idle: waiting for operator start or a higher-level command.
- Scan: acquire and preprocess the point cloud; detect candidate neck regions.

- Plan: synthesize loop pose L and compute an approach trajectory that avoids collisions.
- Launch: spin friction wheels and extrude the loop along the planned trajectory.
- Monitor: observe loop pose and spool tension. A capture is declared if visual enclosure is detected for N consecutive frames or spool tension rises above a capture threshold T_{detect} for a debounce interval $t_{debounce}$.
- Tighten: retract the spool until the target holding tension T_{hold} or a position limit is reached.
- Lift: lift and transport the object while maintaining tension control.
- Release/Recovery: release the object on command or on safety conditions; rewind the loop and return to Idle.

Transitions rely on fused sensor information (point-cloud-based enclosure detection, spool-mounted tension sensor, motor encoder positions) and safety timeouts. For example: if no capture is detected within a maximum launch time t_{max} the controller aborts the attempt and executes a recovery sequence to avoid entanglement.

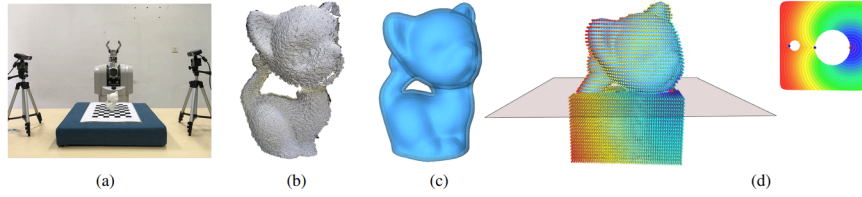


Figure 4.1: Caging-loop calculation algorithm for Lasso Gripper [19]. (a) Setup of the depth reconstruction system. (b) Point cloud captured directly by the depth camera. (c) An r-offset surface built to define the grasping space. (d) Two Morse saddle points (blue) calculated to define the grasping loop.

The algorithm employs a method for identifying the "neck" of the object, which is a critical region for secure gripping. The method entails identifying caging loops, which are closed curves characterized within the embedding space of the target object's shape [19]. By decoupling the caging loops from the surface geometry, the algorithm can synthesize feasible caging grasps that are not constrained by the object's surface details. This enables the gripper to handle objects with complex geometries and topologies effectively.

4.3 Loop Positioning and Tightening

Once the neck region is identified, the algorithm determines the optimal position for the string loop around the neck. The positioning is crucial for ensuring that the loop can be tightened securely without slipping or causing damage to the object. The



Figure 4.2: Illustration of the relative position of the string loop and the target object – an early test

algorithm calculates the precise coordinates for the loop's placement, taking into account the object's shape and the desired gripping force.

The tightening process is executed in a controlled manner, with the string loop gradually constricting around the neck region. The wrist of the gripper approaches the object in synchrony with the tightening action, ensuring that the loop maintains its position and grip. The following pseudocode illustrates the sequence of operations:

Input: point cloud P , string-shooting velocity V

1. Scan P using depth camera
2. Process P to identify key features
3. Identify neck region N in P using the caging loop method
4. Determine optimal loop position L according to V
5. Align lower end of L with N
6. Execute controlled tightening action
7. Approach wrist toward object while maintaining loop position

4.4 Real-Time Feedback and Adjustment

The gripping algorithm incorporates real-time feedback mechanisms to adjust the gripping force and position dynamically. Sensors embedded in the gripper provide continuous data on the tension of the string, the position of the loop, and the force exerted on the object. This data is processed by the controller to make real-time adjustments, ensuring that the grip remains secure and stable.

The force feedback sensor, located on the spool's axle, plays a pivotal role in this process. It monitors the tension of the string during the tightening phase and adjusts the motor's torque accordingly. This prevents excessive force that could damage the object and ensures that the grip is firm enough to hold the object securely.

Chapter 5

Dynamics of the String Loop

Chapter summary

This chapter analyzes the dynamics that govern the formation and stability of the self-supporting string loop used by Lasso Gripper. An analytical model based on a Frenet–Serret description balances tension, gravity, and aerodynamic drag to identify regimes of loop formation and critical junctures in the solution.

Assumptions and simplifications (e.g., negligible string radius, linear drag approximation) are stated explicitly and comparison with representative experimental profiles is used to validate trends. Detailed parametric sensitivity studies and structural FEA are outside this chapter’s scope and are deferred to future work.

5.1 Dynamics of the String Loop

5.1.1 The Dynamics of String Analysis

This section analyzes the loop dynamics that govern Lasso Gripper operation. We focus on the transition from a gravity-dominated regime at low ejection speeds to a self-supporting regime at higher speeds, where aerodynamic drag and inertial effects stabilize the loop shape. The analytic model follows prior theoretical work and is compared with representative experimental profiles to validate qualitative trends [1][6].

At low ejection speeds the loop sags under gravity. As ejection speed increases, aerodynamic drag and inertial forces can balance gravity and internal tension, yielding a self-supporting loop profile. Characterizing these regimes helps select launch speed, loop length, and control strategies that improve capture reliability.

Assuming the loop junction is at position $s = s_0$ when $t = 0$, the position of the junction along this curve is $s = v \cdot t + s_0(\text{mod } L)$. Denote $T(s)$ as the local tension. The Frenet–Serret frame $(\vec{\tau}, \vec{n})$ describes an infinitesimal string element of length ds subjected to its weight $\mu ds \vec{g}$, the local tension $\frac{d(T\vec{\tau})}{ds} ds$, and a linear air drag $-f \vec{\tau} ds$ with $f > 0$.

Newton’s motion equation produces:

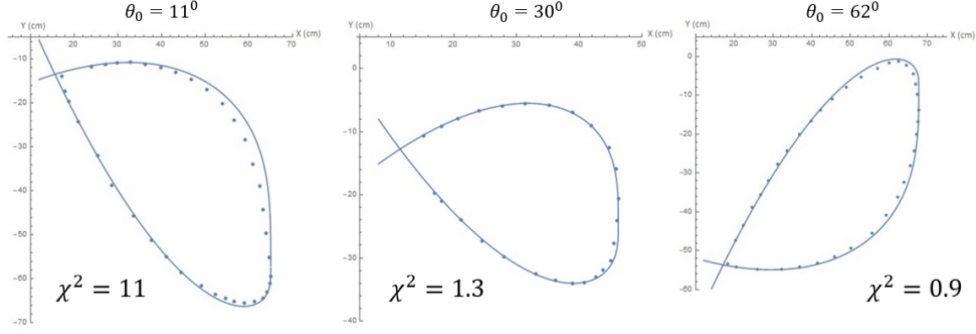


Figure 5.1: Comparison of numerical loop-shape solutions with representative experimental profiles [1]. Example parameters: linear mass $\mu = 0.283 \pm 0.005 \text{ g m}^{-1}$, loop length $L = 160 \text{ cm}$, ejection speed $v = 5.1 \text{ m s}^{-1}$, and several ejection pitch angles θ_0 . A satisfactory fit with $\chi^2 \leq 11$ was obtained for these samples.

$$\frac{d(\mu v \vec{\tau})}{dt} = \mu \vec{g} + \frac{d(T \vec{\tau})}{ds} - f \vec{\tau} \quad (5.1)$$

Noting $ds = v \cdot dt$ and defining the effective tension $T^o = T - \mu v^2$, the following balance equation can be written:

$$T^o \vec{\tau} - f \vec{\tau} + \mu \vec{g} s = \vec{0} \quad (5.2)$$

If the origin is wisely chosen for $\vec{\tau}$ as the point o where the tangent to the loop is vertical, the integration constants will vanish. In the vertical local Frenet–Serret frame, defining $\tan \theta = \frac{dz}{dx}$, the following can be obtained:

$$-f \cdot (x \tan \theta - z) + \mu g s = 0 \quad (5.3)$$

This integro-differential equation admits analytical solutions under the stated simplifications; representative solutions are compared with experiment in Fig. 5.1.

The formation of the self-supporting loop is governed by aerodynamic drag acting on the string. Daerr et al. (2019) investigated this drag both experimentally and theoretically, focusing on a smooth long cylinder moving along its axis [6]. The drag arises from interaction between the moving string and the surrounding air, producing pressure and shear forces that can support the loop.

The loop-shape equations are formulated under the assumption of negligible string radius. They reveal a critical interface separating two regions: a supercritical zone (string speed larger than wave speed) and a subcritical zone (wave speed larger than local string speed). Regular solutions that cross this juncture continuously can be obtained and compared with experimental profiles.

In practical terms, understanding the string loop dynamics has significant implications for Lasso Gripper’s design and operation. By leveraging the principles of aerodynamic drag and the transition between different velocity regimes, Lasso Gripper can achieve precise control over the string loop’s shape and stability. This enables the gripper to handle objects with varying sizes and shapes effectively, improving robustness of capture.

5.2 Workspace Analysis

Due to the self-supporting loop configuration of Lasso Gripper in three-dimensional space, the operational workspace of the proposed manipulator significantly exceeds the physical reach of the robotic arm.

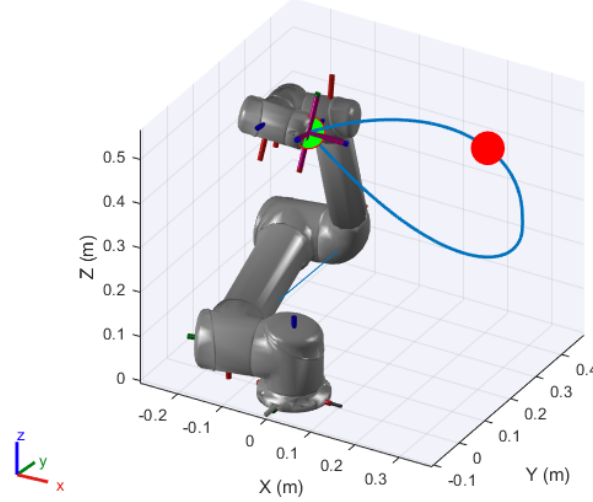


Figure 5.2: Lasso Gripper string loop is set at the end of UR5 robotic arm. The green point at the end of the robotic arm marks the position reachable by a rigid manipulator, while the red point indicates the furthest position on the string loop from the original point at the same robotic position.

As delineated in Section 5.1, the three-dimensional pose kinematics of the lasso-type end-effector can be characterized by constrained dynamics under gravitational loading. For the local coordinate system constrained by gravity, the working plane is defined as the plane formed by the lasso’s ejection direction and the gravity direction. The rotation with the normal vector of this working plane as the rotation axis is defined as pitch, while the rotation with the gravity vector as the rotation axis is defined as yaw. According to the right-hand coordinate system, the roll rotation axis is defined as the axis orthogonal to both the pitch and yaw rotation axes, lying within the working plane. Specifically, the gripper mechanism exhibits negligible roll angular displacement within the global coordinate frame due to gravitational stabilization. Its pitch orientation

becomes parametrically dependent on the projectile's ejection velocity vector, while yaw angle variations are governed by the robotic manipulator's pose in Cartesian space.

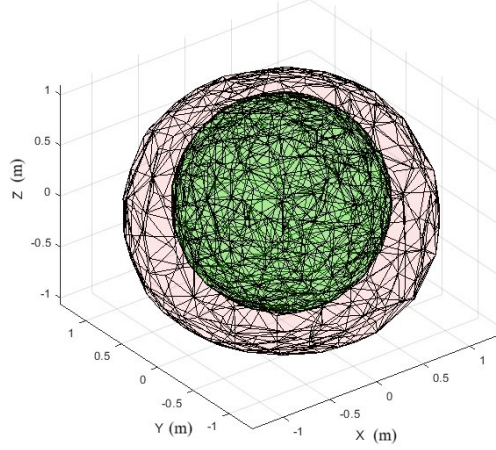


Figure 5.3: The initial robotic arm workspace is marked in green, and the extended Lasso Gripper workspace is marked in red.

A case study is conducted with specific parameters to investigate the workspace of Lasso Gripper on a 6-axis robotic arm using the Monte Carlo method. A UR5 robot without joint angle limits is used as an example in Fig. 5.2, and $[R, x_+, x_-]$ is set as $[0.7, 0.2\text{ m}, 0.1\text{ m}]$ in this case. The green point at the end of the robotic arm is the specific position that a rigid manipulator can reach, and the red point represents the furthest point on the string loop from the same robotic position. The sets of green and red points in Fig. 5.3 are the workspace of the robotic arm and the extended workspace of Lasso Gripper, respectively. After generating 200,000 sampled robotic-arm configurations, the convex-hull workspace volume is calculated. The volume of the original UR5 workspace convex hull is 3.4620 m^3 , and the volume of the extended Lasso Gripper workspace convex hull is 8.9002 m^3 . The workspace extension ratio reaches 157.08%.

Chapter 6

Experimental Validation (Qualitative Demonstrations)

6.1 Objectives

To validate the unique capabilities of Lasso Gripper and to compare its performance with antipodal grippers, the following experimental procedure was designed. The setup employed a modular robotic arm (Inovo Robotic) with a maximum payload of 6 kg, mounted on a flat platform where the target objects were placed. The experimental objectives were divided into four categories:

1. Capturing animal figures by the neck or horns,
2. Capturing objects of diverse shapes,
3. Capturing oversized objects through shape adaptation, and
4. Capturing moving objects.

6.2 Experimental Setup

For Objectives 1 to 3, the procedure consisted of the following steps:

1. The robotic arm moved to a predefined position to initiate the launch phase;
2. Lasso Gripper was swept toward the target while maintaining its open-loop configuration;
3. Upon reaching the target, the string was retracted to secure the object;
4. The arm performed a vertical lifting motion while maintaining a stable grip.

For Objective 4, the robotic arm remained stationary, and Lasso Gripper independently executed the capture process aimed at intercepting a moving object in flight.

6.3 Experimental Procedure

The experimental procedure was divided into phases to evaluate Lasso Gripper's performance:

1. **Static Object Gripping Tests:** Static object gripping tests were conducted to assess the gripper's ability to handle stationary objects. Test objects were placed in fixed positions, and the gripping algorithm was executed to secure and lift each object. Success rates, gripping forces, and any slippage or deformation of the objects were recorded where feasible.
2. **Dynamic Object Gripping Tests:** Dynamic object gripping tests focused on evaluating the gripper's performance with moving objects. Test objects with controlled motion, such as rolling or sliding trajectories, were introduced and the lasso attempted to intercept and secure them. The effectiveness and speed of the gripping action were measured to assess performance in dynamic scenarios.
3. **Comparative Analysis:** Finally, a comparative analysis was conducted to compare Lasso Gripper's performance with traditional gripping methods. Parallel tests using a standard parallel-jaw gripper were carried out on the same set of test objects. Differences in gripping force, handling precision, success rates, and overall efficiency were analyzed to highlight advantages and potential areas for improvement of Lasso Gripper.

6.4 Data Collection and Analysis

Data collection involved capturing both quantitative and qualitative metrics where available. Key measurements included gripping force (recorded via force sensors) and positioning accuracy (assessed with motion tracking to evaluate loop placement around candidate neck regions). Where feasible, counts of successful and failed grasps were noted; however, the evaluation emphasis in this chapter is demonstrative rather than a full statistical study. Object integrity was inspected for visible damage or deformation after handling, and common failure modes were logged to guide subsequent design iterations.

Limited summary statistics were computed where appropriate, but a comprehensive statistical campaign (large-sample success rates, fatigue tests) was outside the present scope and is recommended for future work.

6.5 Results and Discussion

The evaluation reported in this chapter is demonstration-focused and intended to validate feasibility, illustrate common operating modes, and highlight comparative behavior versus conventional antipodal grasping. Test cases emphasized scenarios where

a loop-based capture offers clear mechanical advantages. Results are presented as illustrative sequences and representative frames rather than as full statistical studies. Observed failure modes (entanglement, missed captures at extreme approach angles, and tension-management lapses) are documented to guide design improvements.

6.5.1 Various Objects Grasping Results

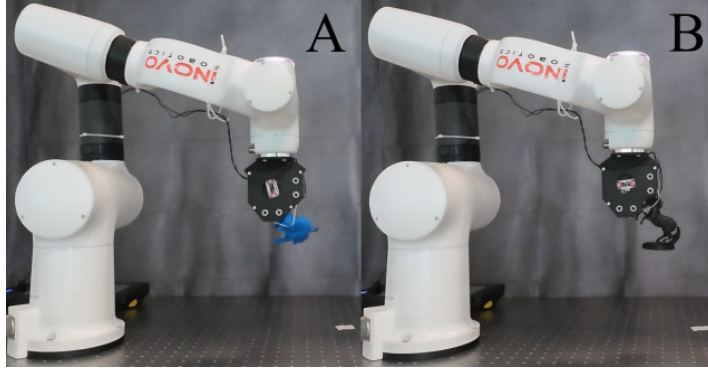


Figure 6.1: Validation of grasping ability via bull and horse figures.

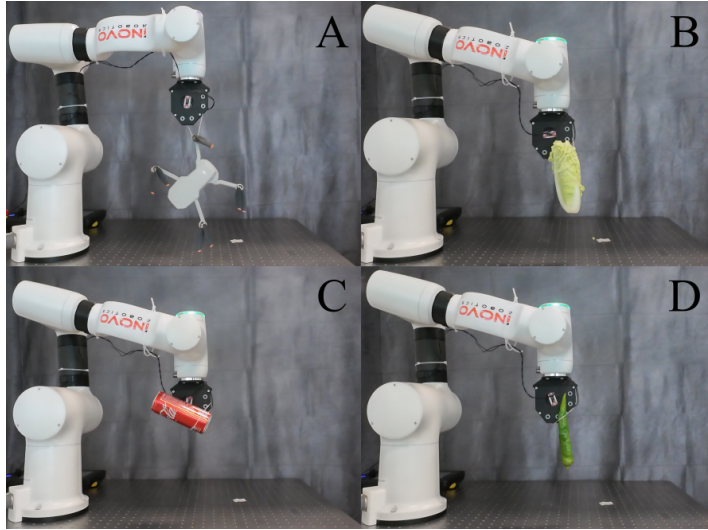


Figure 6.2: Validation of grasping ability over variable shapes: drone, cabbage, coke can and chili pepper.

Fig. 6.1 illustrates the experimental results for Objective 1. The lasso attached to the horns and necks and secured the figures, allowing the robot to lift and transport them from the platform. For the second set of target objects (a drone, a cabbage, an empty coke can, and a chili pepper), Lasso Gripper successfully grasped the targets and carried them away from the base. The grasping strategy for these experiments was similar. In both cases, the robot swept above the target while Lasso Gripper maintained the string

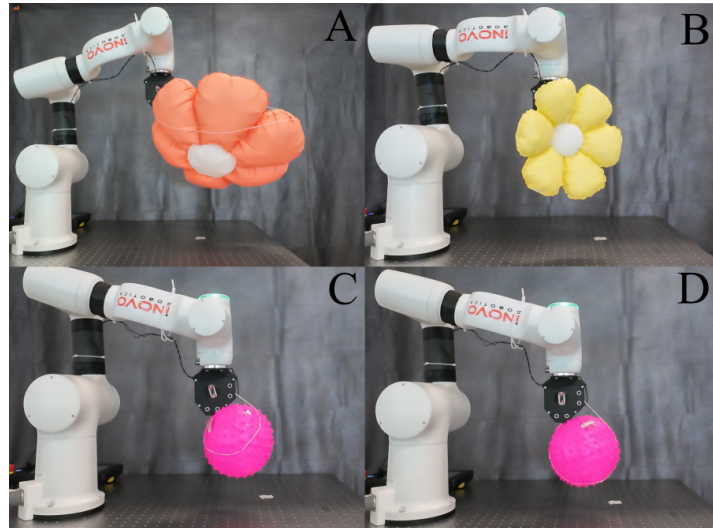


Figure 6.3: Validation of grasping ability with oversized objects: inflated balloons.

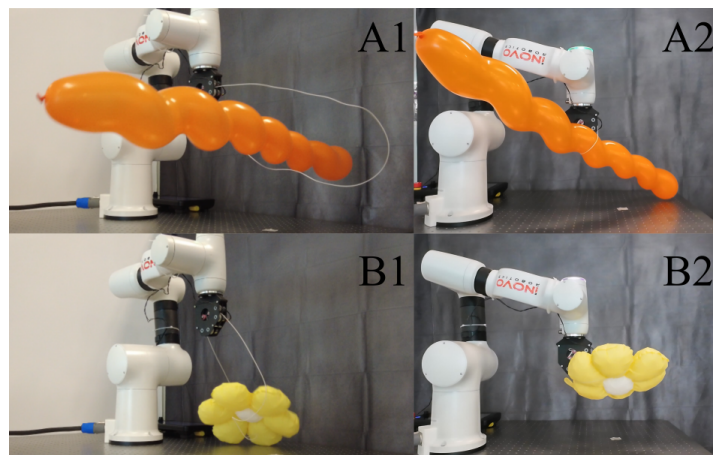


Figure 6.4: Flying object is captured by Lasso Gripper.

in its full range, ensuring that the target could be captured as the string retracted. This strategy was extended to expansive targets such as oversized balloons. The third set comprised two flower-shaped balloons (diameters 45 cm and 71 cm) and a spike ball of 18 cm diameter, as shown in Fig. 6.3. Although Lasso Gripper could not fully encircle the largest balloons due to loop-size limitations, the loop still secured them sufficiently to lift them from the platform. This observation led to the fourth case, capturing flying objects. Fig. 6.4 shows examples where a long balloon and a small flower balloon were manually thrown through Lasso Gripper’s loop and successfully intercepted.

Some notable phenomena were observed during the demonstrations, illustrating additional advantages of Lasso Gripper. The string’s sustained high speed made it possible to grasp objects placed close to surfaces. In Fig. 6.2 (B) and (D), where the cabbage and chili pepper were placed near the table, Lasso Gripper maintained the loop shape and achieved successful grasps. The adaptive, distributed contact provided by the loop tolerated deviations in placement; compared with the grasp posture in Fig. 6.3 (D), the loop in Fig. 6.3 (C) exhibited an unexpected pose deviation yet still produced a secure grip. Thus, Lasso Gripper demonstrates adaptive force-closure behavior.

6.5.2 Comparisons with Antipodal Point Gripper

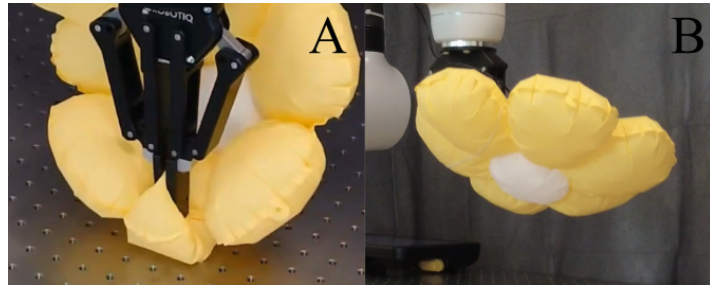


Figure 6.5: The inflated balloon is captured by the antipodal gripper. As a result of the concentrated stress applied, the balloon was severely deformed.

As the comparison group, the 2F-140 gripper from Robotiq was used as the classical antipodal gripper. In this scenario, the small flower balloon served as the target due to its elasticity. As shown in Fig. 6.5, the flower balloon experienced severe deformation when grasped by the antipodal gripper. Although the balloon remained intact after the trial, the limitation of traditional grippers for delicate or highly deformable objects was evident. Instead of concentrating forces on small contact patches, Lasso Gripper captures objects via distributed rope tension, avoiding damage to delicate surfaces and enabling handling of larger, potentially heavier objects than some traditional grippers.

Taken together, these demonstrations validate the feasibility of Lasso Gripper as both a novel end-effector and a mechanism study platform for tendon-driven manipulation. The results show that controlled tension, distributed contact, and low-distal-complexity actuation can produce robust behavior under uncertainty in object size, shape, and motion. The next part of this thesis transfers these mechanism insights to

a more constrained setting, where the same tendon-driven principles must satisfy the stricter requirements of handheld surgical instrumentation.

Chapter 7

Handheld Tendon-Driven Surgical Instrument: Design, Packaging and Control

Chapter summary

This chapter presents a translational embodiment of the tendon-driven mechanism principles developed earlier: a compact, handheld laparoscopic instrument. It first summarizes the primary design drivers—low distal mass, ergonomic balance, repeatable bidirectional tension transmission, and safety—and describes the overall system architecture linking proximal actuation, paired-cable reels, and quick-swap end-effectors. The chapter then details the mechanical implementation (actuator selection, cable routing and guidance, paired-cable reels, tip interfaces and packaging) and the control/HMI stack (encoder-based intent mapping, low-level tension and position loops, firmware communication, and homing/safety procedures). Emphasis is placed on packaging and routing trade-offs that preserve tension and enable clinically useful distal dexterity while keeping the device practical for handheld use.

7.1 Necessity of Developing a Handheld Motorized Medical Instrument

The preceding chapters established a set of mechanism-level insights from Lasso Gripper: tendon-driven systems can achieve adaptive interaction through controlled tension, can reduce distal inertia through proximal actuation, and can generate rich behavior through careful routing of flexible tensile elements. The next question is whether these same principles remain effective when transferred to a domain with far tighter geometric, ergonomic, and safety constraints. Handheld laparoscopic instrumentation provides exactly such a testbed.

Contemporary research in surgical robotics has predominantly prioritized maximizing tip-positioning accuracy through advanced actuation methods and the integration of supplementary sensors. While these approaches demonstrably enhance precision, they often overlook a critical practical constraint: the weight-to-functionality ratio. The evident correlation between enhanced capabilities achieved through more powerful motors or additional sensing and increased instrument mass creates a significant adoption barrier. This added weight often leads surgeons to favor simpler, nearly weightless traditional instruments over their motorized counterparts, despite the latter’s technological advantages.

To resolve this fundamental trade-off, the present work introduces a holistic approach to ergonomic and functional design in robotic-assisted surgery. This chapter reframes the surgical instrument as a second embodiment of the tendon-driven mechanism thesis: instead of maximizing grasp adaptivity as in Lasso Gripper, the objective here is to maximize distal dexterity and control fidelity while preserving low mass and practical usability. The proposed system embodies a dedicated effort to distribute system components intelligently, minimize inertial load, and integrate multi-modal sensing without compromising the manipulator’s agility or the surgeon’s comfort. In this sense, the chapter marks a transition from open-environment adaptive grasping to clinically constrained tendon-driven manipulation.

7.2 Core Design Advantages for Handheld Applications

This cable-driven, differential actuation philosophy offers several fundamental advantages that are particularly suited to the development of advanced handheld devices:

- **Enhanced Dexterity and Manipulation:** The system enables multi-degree-of-freedom manipulation (e.g., rotation, twisting) using a minimal number of actuated elements. This is crucial for applications requiring fine motor control within confined spaces.
- **Inherent Compliance and Safety:** The system is naturally compliant. Cables under tension provide a direct force-feedback path. If an unexpected obstacle is encountered, the cables will slacken rather than apply a rigid, potentially damaging force. This passive safety feature is a critical asset for interactions in unstructured or delicate environments.
- **Proximal Actuation and Minimal End-Effector Mass:** A primary benefit for handheld device design is the ability to locate the heavy actuators (motors) and associated drive electronics proximally, either in the device’s handle or even in a remote unit. This dramatically reduces the mass and inertia of the distal end-effector, leading to improved ergonomics, reduced user fatigue, and the potential for higher maneuverability and smaller form factors at the interaction point.

7.3 Defining Performance Metrics for Structural Design

To translate this conceptual framework into a viable product, key performance metrics must be established. These metrics are the surgical-domain counterpart of the mechanism criteria discussed in Lasso Gripper chapters: instead of capture range and adaptive loop closure, the emphasis now shifts to precision, low hysteresis, compact routing, and stable bidirectional tension transmission. They will directly guide the structural design choices detailed in the next chapter:

- **Precision and Resolution:** The minimum achievable incremental movement of the end-effector tip, targeting sub-millimeter and sub-degree accuracy for delicate tasks.
- **Force Capability:** The maximum tensile force the cables must transmit to securely grasp and manipulate objects of a target weight range.
- **Range of Motion:** The required angular deflection and rotational range of the end-effector to perform its intended functions effectively.
- **Force Transmission Efficiency:** Minimizing friction losses throughout the cable drive path is essential for accurate force feedback and predictable control.
- **Low Hysteresis and Backlash:** The structural design must ensure minimal play and elastic deformation in the system to guarantee precise, repeatable, and immediate response to actuator inputs.

7.4 Pathway to Structural Implementation

This chapter has outlined how the tendon-driven principles distilled from Lasso Gripper can be reformulated for a handheld surgical device. The core principles of compliance, proximal actuation, differential control, and explicit tension management remain the same, but their implementation must now satisfy stricter constraints on packaging, repeatability, and operator ergonomics.

The next chapter addresses the critical task of transforming this mechanism framework into a robust mechanical reality. The structural design must meticulously integrate several key subsystems:

1. **Actuation Unit Selection and Layout:** Choosing motors and sensors that meet the performance metrics and determining their optimal arrangement.
2. **Cable Routing and Guidance System:** Designing low-friction, precise pathways for the cables to ensure efficient force transmission and longevity.
3. **End-Effector Configuration and Cable Termination:** Engineering the interface where the cables attach to the manipulator, ensuring secure anchoring, precise motion, and mechanical advantage.

Through this structured approach, the subsequent design will show how a mechanism concept first explored through adaptive loop grasping can be translated into a functional, high-performance cable-driven surgical instrument.

The design choices described in this chapter were strongly influenced by the earlier development of Lasso Gripper. Rather than treating the handheld instrument as an isolated engineering project, this thesis positions it as a second embodiment that tests and extends mechanism-level hypotheses first clarified through Lasso Gripper: that tensile elements can create useful compliant interaction, that differential tendon routing can produce compact multi-DOF behavior, and that proximal actuation can preserve low distal inertia. The difference is not in the underlying mechanism logic, but in the operating constraints. Lasso Gripper validates tendon-driven adaptivity in a relatively open manipulation setting, whereas the surgical instrument validates tendon-driven dexterity and controllability in a tightly constrained medical setting. Emphasizing this continuity keeps the thesis focused on tendon-driven mechanisms rather than on two unrelated devices.

Chapter 8

Mechanical Implementation of the Handheld Instrument

Chapter summary

This chapter describes the mechanical and electromechanical implementation of the handheld tendon-driven instrument. It covers actuator selection and arrangement, cable routing and guidance, paired-cable reels and winding mechanisms, tip design and quick-swap interfaces, input module architecture, and packaging considerations for preserving tension, minimizing hysteresis, and reducing distal mass.

8.1 Mechanism and Design

8.1.1 Output Motors

A key challenge for handheld cable-driven devices is selecting actuators that combine low mass with sufficient torque. The chosen output actuator, the Feetech STS3032 smart servo, offers a favorable performance-to-weight ratio and built-in feedback useful for closed-loop control. It weighs approximately 9 g and provides a stall torque of about $4.5 \text{ kg} \cdot \text{cm}$, enabling reliable tensioning through small-diameter pulleys.

The STS3032's serial-bus communication simplifies wiring and supports daisy-chaining, which reduces the number of conductors running through the handle and aids miniaturization. As a smart servo it reports position, speed, and load, allowing the controller to perform basic load estimation and to implement tension-aware control strategies that improve positional accuracy and protect against overloading.

8.1.2 Input Motors and Sensors

The control philosophy for this cable-driven handheld device is based on replicating the user's hand movements with high fidelity and translating them into precise motions of the distal end-effector. This requires a sophisticated input system that can accurately capture the intent of the user's fingers while providing sufficient force feedback and stability.



(a) Feetech 3032 for output



(b) Feetech 3009 for wrist inputs

Figure 8.1: Feetech servos: left motor for output (STS3032), right motor for wrist inputs (STS3009)

For the input actuation, shown in Fig.8.1(a), the Feetech STS3009 servo was selected based on a different set of requirements. Weighing 40 grams and providing a stall torque of 0.3 Nm, this more powerful motor serves two essential functions in the input stage: it maintains mechanical equilibrium by resisting forces from the output tendons, thus offering a stable and responsive feel to the operator, and it helps filter out high-frequency physiological tremors or unintended jerks, ensuring that only deliberate commands are transmitted to the end-effector.

However, since the two input motors alone cannot fully capture all nuanced finger movements, a hybrid sensing strategy was implemented. Two AMS AS5600 magnetic rotary encoders were integrated into the input mechanism to directly and accurately detect the user's intent. These sensors provide ultra-high 12-bit resolution and contactless sensing, enabling precise recognition of two distinct actions: the pinching gesture of the index finger and thumb, which controls grasping, and the rotational movement of the fingers, which is translated into in-plane rotation of the end-effector. This combination of robust actuation and high-resolution sensing creates an intuitive and stable human-machine interface suitable for delicate manipulation tasks.

8.2 Cable Routing and Guidance System

8.2.1 Tip Design

In addition to motor performance, the accuracy and repeatability of the end-effector's motion are highly dependent on precise cable guidance and sustained tension control. To ensure consistent behavior during operation, each cable must be constrained to move freely yet securely within a meticulously designed path, minimizing unwanted friction and slack.

A specialized tip assembly has been developed to fulfill these requirements. As shown in Fig.8.2, the four cables—grouped into two pairs, each controlling one jaw of the

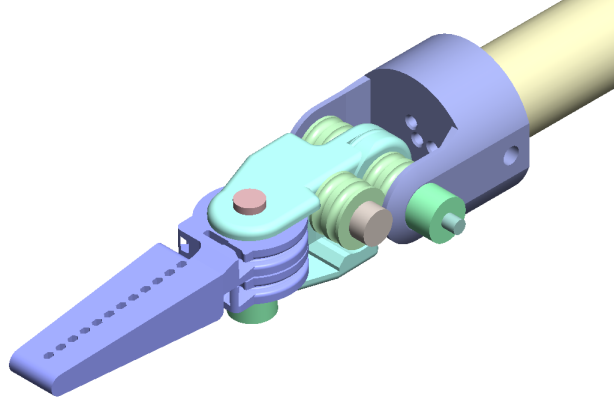


Figure 8.2: Tip Design for Cable Guidance

pliers—are routed through custom-machined channels. These channels ensure that the cables remain aligned throughout the entire range of motion, avoiding cross-interference and reducing wear.

The two jaw components (fabricated in blue) are mounted onto a central middle base (green). To further enhance cable stability and minimize friction, donut-shaped guides machined via CNC (light green) are integrated into the middle base. These guides act as low-friction directional pulleys, maintaining cable position with high precision and reducing bending stress, thereby extending cable service life.

The main base housing (deep blue) is designed with precision-aligned holes that direct the cables perpendicularly toward the motors. This alignment is critical for maximizing force transmission efficiency and eliminating off-axis loading, which can introduce positioning errors and reduce system responsiveness.

Through this integrated approach—combining dedicated guiding channels, spherical contact points, and perpendicular alignment—the system achieves high repeatability, minimal hysteresis, and smooth force feedback, making it suitable for delicate manipulation tasks requiring fine motor control.

Finite Element Analysis of Tip Components

To validate the structural integrity of the tip assembly, finite element analysis (FEA) simulations were conducted on the two primary load-bearing components: the jaw (gripper) and the middle base. Both models used Aluminum Alloy 6061 as the material. A mesh size of 0.2 mm was adopted and the pivot shaft was modeled as a fixed support. Cable contact forces were applied as distributed loads over a circular area of 1.2 mm diameter to represent the cable-guide contact.

Jaw (gripper) FEA Model setup: mesh size 0.2 mm, material Al6061, pivot fixed. A 6 N load was applied over a 1.2 mm diameter circular contact area at the cable interface. Results show a maximum von Mises stress of 12.4 MPa (see Fig. 8.3). This stress level is

well below the typical yield strength of 6061 aluminum (≈ 55.1 MPa), indicating a large safety margin for the jaw under the applied loading.

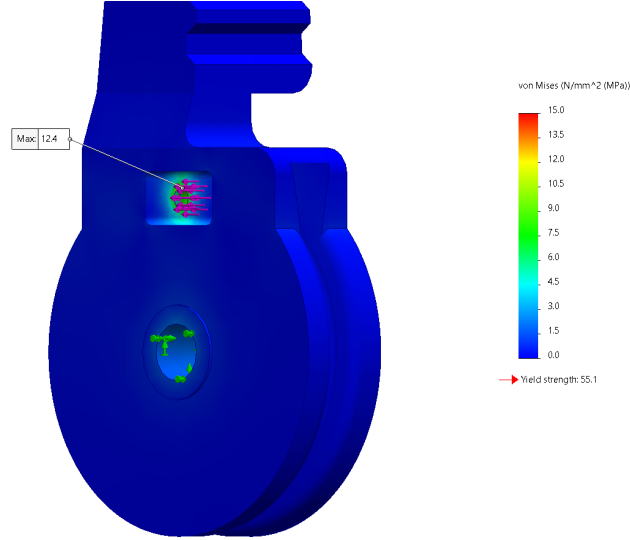


Figure 8.3: Jaw FEA: von Mises stress distribution under a 6 N cable load

Middle base FEA Model setup: mesh size 0.2 mm, material Al6061, pivot fixed. A 6.5 N load was applied over a 1.2 mm diameter circular contact area at the cable interface. In addition, the transmitted loads from a pair of jaws (two 6 N forces) were applied to the upper pivot to represent the combined loading path. The analysis yields a maximum von Mises stress of 9.1 MPa (see Fig. 8.4), also well below the material yield strength.

Overall, the FEA results indicate that both the jaw and middle base have ample structural capacity under the stated loading conditions. Local stress concentrations near cable holes and sharp edges were identified; applying small fillets, ensuring adequate material thickness around guide features, and standard deburring practices are recommended to mitigate these effects.

8.2.2 Reel

A fundamental challenge in cable-driven mechanisms lies in maintaining adequate tension during both winding and unwinding phases. In a conventional single-motor-single-cable configuration, unidirectional actuation presents a significant limitation: while rotation in one direction (e.g., clockwise) effectively pulls and tensions the cable, reverse rotation (counterclockwise) does not positively push the cable but often results in slack accumulation along the path. This leads to unpredictable response, reduced positional accuracy, and potential entanglement.

To address this issue, a paired-cable design has been implemented. The two cables corresponding to a single degree of freedom are affixed to the same axial shaft driven by one motor. This configuration effectively reforms the two independent cables into

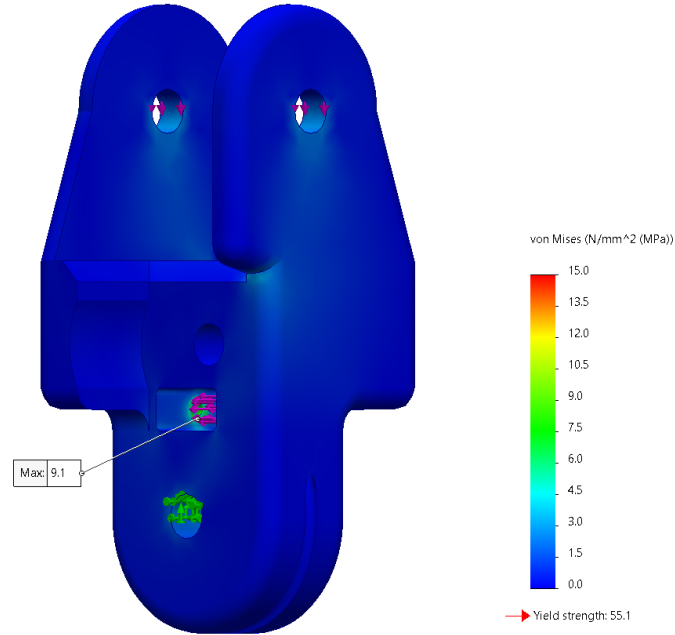


Figure 8.4: Middle base FEA: von Mises stress distribution under combined cable and jaw-transmitted loads

a continuous loop—functioning conceptually as a flexible drive belt. As a result, when the motor rotates in one direction, one cable is pulled while the other is simultaneously recovered, enabling bidirectional force transmission without slack accumulation.

The key to ensuring effective operation of this belt-like linkage is sustained cable tension. To achieve this, the reel assembly is split into two distinct components, each dedicated to one cable. These two reel pieces are mounted on the same shaft and are designed to rotate in opposite directions relative to each other. This counter-rotation ensures that one cable is wound while the other is unwound during actuation, maintaining constant tension across the system.

To prevent slippage and undesired loosening of the cables during operation, one-way bearings are incorporated into the reel mechanism. As the name implies, these bearings permit free rotation in one direction while locking in the opposite direction. This feature is critical for preserving tension integrity, especially during transitions in motor direction or under varying load conditions. The use of one-way bearings enhances system reliability, improves motion accuracy, and minimizes the need for manual tension adjustments.

8.2.3 Motor Arrangement and Quick-Swap Mechanism

As previously discussed, the STS3032 servos are employed as the driving motors for the cable reels. In practical surgical scenarios, various end-effector tips are often required throughout a single procedure. To accommodate this need, a quick-swap system has

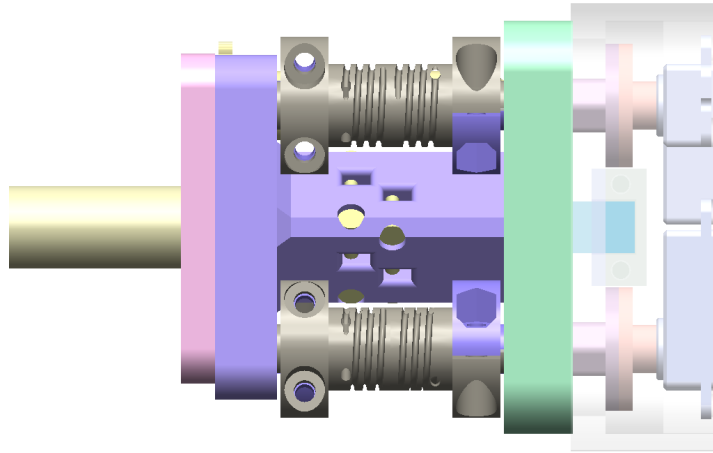


Figure 8.5: Reel Design and Winding Mechanism

been integrated into the design, allowing for rapid and secure tip replacement without compromising system integrity.

To achieve a compact form factor suitable for handheld operation, the four motors are arranged in a rotationally symmetric pattern within the main housing. This layout optimizes spatial efficiency while maintaining sufficient clearance for motor power and communication cables, thereby preventing interference and ensuring reliable connectivity. The quick-swap mechanism incorporates a user-friendly yet secure attachment

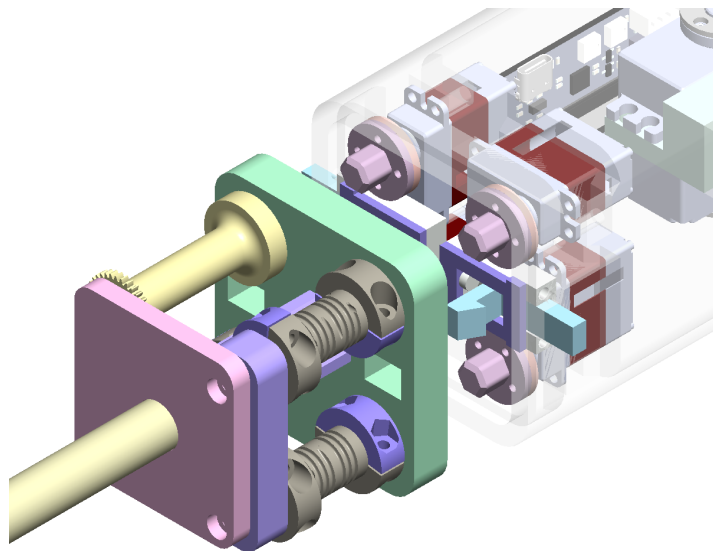


Figure 8.6: Motor Arrangement and Quick-Swap Mechanism

scheme. Press-activated hooks are mounted within the main frame, enabling tool-free

engagement and release. To attach a tip module, the user simply aligns and applies gentle pressure until the hooks audibly lock into place. For disengagement, bilateral release buttons must be depressed simultaneously while withdrawing the module. This dual-action release strategy prevents accidental detachment during operation, enhancing safety, while minimizing the number of steps required for module exchange.

This design ensures a robust mechanical and electrical connection during use, maintaining precise cable alignment and tension integrity while facilitating efficient tool interchange in time-sensitive surgical environments.

8.2.4 Input Design

The input module is designed to intuitively capture the user’s hand and finger motions with high precision and minimal latency. By integrating two STS3009 servos and two AS5600 magnetic encoders, the system is capable of sensing four degrees of freedom (DOFs): two DOFs from the wrist movement—yaw and pitch—and two DOFs from the fingers, namely roll and open/close actuation.

The wrist motions are translated through a dual-axis gimbal-like mechanism, where the STS3009 servos provide both actuation force and preliminary positional feedback. Fine rotational movements in yaw and pitch are accurately detected and relayed to the control system, enabling responsive and natural manipulation of the end-effector. Finger actions are captured using the high-resolution AS5600 encoders, which detect

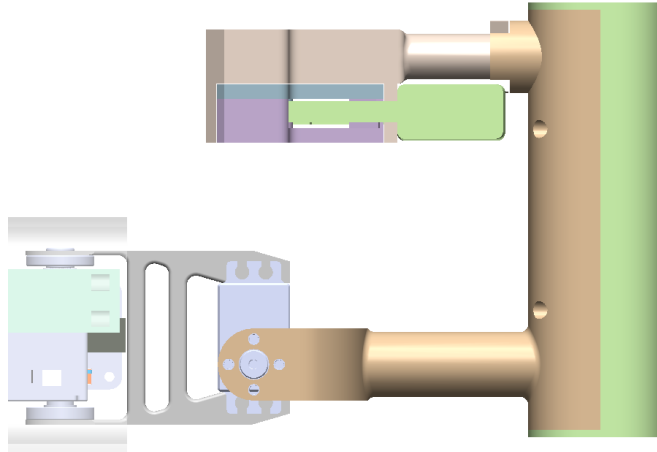


Figure 8.7: Input Module Design

both the rolling gesture and the degree of opening or closing of the fingers. Specifically, the open/close mechanism is implemented as an interchangeable module, allowing customization based on user preference or task requirements. The lever length and rebound force can be easily adjusted or swapped, facilitating ergonomic adaptability and reducing user fatigue during prolonged operations.

This modular and sensor-rich input design ensures that the system can accommodate a wide range of hand sizes and operating styles while maintaining accurate and reliable motion mapping between the user and the instrument.

Chapter 9

Control Strategy and Implementation

Chapter summary

This chapter presents the control architecture for the handheld instrument. It includes the modeling approach, Denavit–Hartenberg frames and forward kinematics, system architecture and data flow, intent estimation and command mapping, low-level actuation and tension management, safety and homing procedures, and firmware communication and human–machine interface considerations.

9.1 Control Strategy and Implementation

9.1.1 Modeling

The project aims to implement a da Vinci–style surgical instrument claw effect using a compact 4-DOF hardware architecture: a 3-DOF manipulator arm plus a single gripper open/close degree of freedom. The system input is provided by the handle as three rotational joint angles and one gripper opening angle.

Since the manipulator itself has only three actual mechanical degrees of freedom, it cannot realize an arbitrary rotation axis in space. A specified direction in space requires three independent pose parameters, and a stable pose also consumes three degrees of freedom. This makes the system inherently over-constrained if arbitrary axis rotation is demanded. In contrast, the da Vinci system achieves effective 6-DOF stabilized control by using a larger arm to drive a smaller wristed arm in coordinated motion. Under the hardware constraints of this project, that six-degree-of-freedom stabilization strategy is not considered.

Instead, the control design adopts a decoupled attitude mapping. The coordinate system is defined using Euler angles to describe the desired end-effector orientation. The input attitude from the handle is mapped linearly to the output attitude of the manipulator, subject to the arm’s three-degree-of-freedom limitations.

9.1.2 Denavit–Hartenberg Coordinate Frames

The coordinate frames are established using standard Denavit–Hartenberg conventions, as shown in Figures 9.1 and 9.2. The manipulator consists of three rotary joints, with the following D-H parameters:

Table 9.1: Denavit–Hartenberg parameters for the 3-DoF manipulator

Link	θ_i	d_i	a_i	α_i	Remarks
1	θ_1	0	0	-90°	Base rotation
2	$90^\circ - \theta_2$	0	$-a_2$	90°	Upper arm rotation
3	$\theta_3 - 180^\circ$	0	a_3	0	

The following rotation matrix describes the end-effector orientation:

$$A_1 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} s_2 & 0 & c_2 & -a_2 s_2 \\ c_2 & 0 & -s_2 & -a_2 c_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_3 = \begin{bmatrix} -c_3 & -s_3 & 0 & -a_3 c_3 \\ s_3 & -c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The end-effector pose is governed by the following equations:

$$\begin{aligned} P_x &= c_1 s_2 (-a_3 c_3) + (-s_1)(a_3 s_3) + c_1 c_2 (0) + (-a_2 c_1 s_2) \\ &\rightarrow P_x = -a_3 c_1 s_2 c_3 - a_3 s_1 s_3 - a_2 c_1 s_2 \\ P_y &= s_1 s_2 (-a_3 c_3) + c_1 (a_3 s_3) + s_1 c_2 (0) + (-a_2 s_1 s_2) \\ &\rightarrow P_y = -a_3 s_1 s_2 c_3 + a_3 c_1 s_3 - a_2 s_1 s_2 \\ P_z &= -c_2 (-a_3 c_3) + 0(a_3 s_3) + s_2 (0) + a_2 c_2 \\ &\rightarrow P_z = a_3 c_2 c_3 + a_2 c_2 \end{aligned}$$

The orientation of the end effector can also be expressed through the base axis direction vectors shown in Table 9.2.

Table 9.2: End-effector base-axis direction vectors

Direction	X-axis (n)	Y-axis (o)	Z-axis (a)
Base X-axis	$-c_1s_2c_3 - s_1s_3$	$-c_1s_2s_3 + s_1c_3$	c_1c_2
Base Y-axis	$-s_1s_2c_3 + c_1s_3$	$-s_1s_2s_3 - c_1c_3$	s_1c_2
Base Z-axis	c_2c_3	c_2s_3	s_2

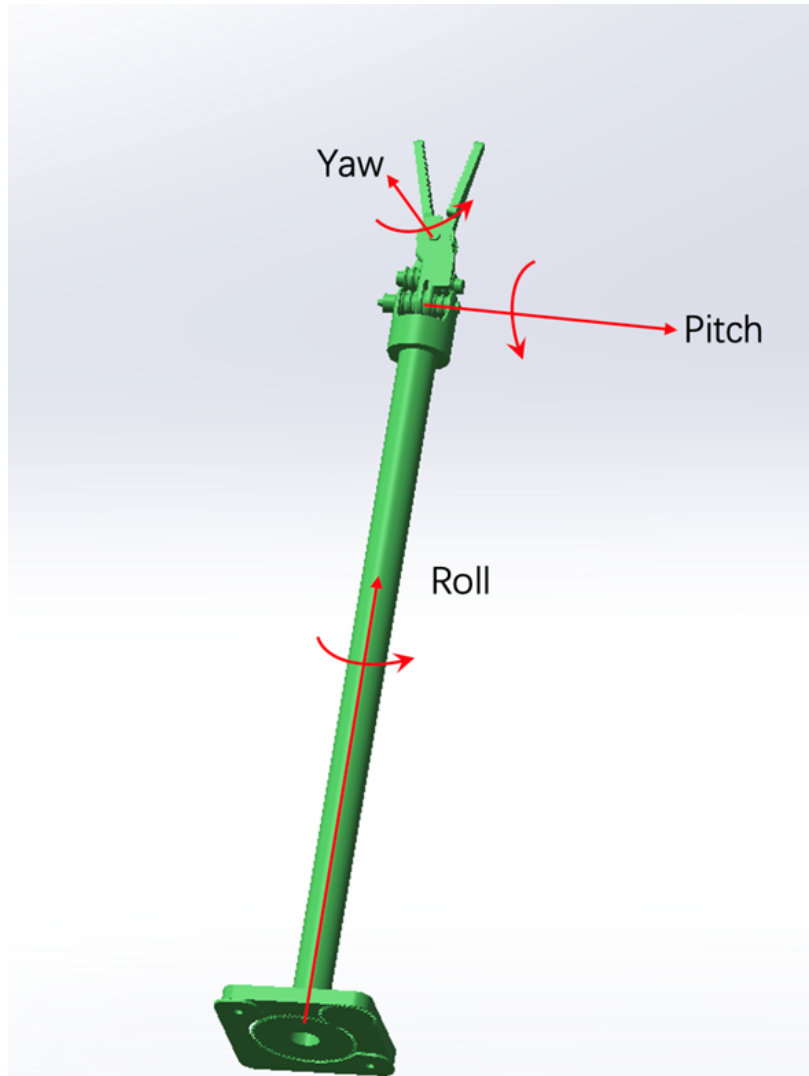


Figure 9.1: Denavit–Hartenberg coordinate frame definition for the manipulator

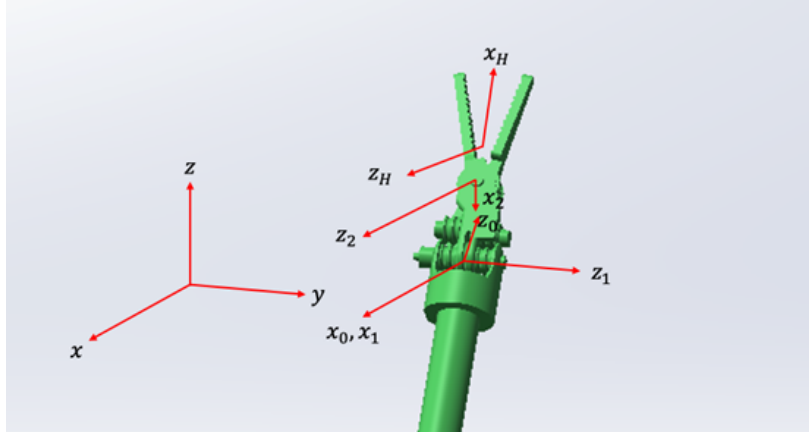


Figure 9.2: Alternative view of the D-H coordinate frames and joint axes

9.1.3 System Architecture and Data Flow

The control system is implemented on an ESP32 microcontroller running FreeRTOS, enabling real-time processing of sensor inputs and actuator commands. The architecture follows a cyclic loop structure, as illustrated in the firmware code, where sensor feedback is continuously acquired, processed, and translated into motor commands.

At the core of the system is a feedback loop that reads position, speed, load, voltage, temperature, and current from eight Feetech STS-series servos via UART communication. This multi-modal feedback provides the foundation for state estimation and control. The input module captures user intent through four degrees of freedom: wrist yaw, pitch, roll, and gripper opening/closing, using AS5600 magnetic encoders and STS3009 servos for precise motion detection.

9.1.4 Intent Estimation and Command Mapping

User inputs are mapped from encoder readings to angular commands. The firmware calculates roll, pitch, yaw, and gripper commands by scaling encoder positions relative to a midpoint. For instance, $roll_{cmd}$ is derived from the difference between encoder p8 and the midpoint, scaled to degrees. These commands are then transformed into servo-specific inputs, accounting for mechanical ratios and phase offsets to ensure coordinated motion.

The transformation incorporates kinematic considerations, such as the differential yaw control for the gripper jaws, where $s3_yaw_right$ and $s4_yaw_left$ are adjusted based on yaw_in and $gripper_cmd$ to achieve symmetric or asymmetric grasping.

To robustly infer operator intent in the presence of noise and delay, an Extended Kalman Filter (EKF) can be integrated as a future enhancement. The EKF would fuse sensor inputs from AS5600 encoders, motor encoders, and optional inertial measurements from an IMU module. It maintains estimates of finger poses, wrist orientation,

and cable tensions, enabling predictive control and improved stability during rapid transitions. This probabilistic approach handles uncertainties in sensor readings, providing smoother and more reliable command generation compared to the current direct mapping.

9.1.5 Low-Level Actuation and Tension Management

Actuation is handled through direct position control of the STS3032 servos for the output stage and STS3009 for input stabilization. The code implements a locking mechanism for the gripper motors: when the gripper is open beyond a threshold ($gripper_{open} = 5.0degrees$), the motors are unloaded to reduce power consumption and allow free motion; otherwise, they are locked to maintain tension.

This tension management is critical for cable-driven systems, preventing slack while enabling compliant interaction. The firmware uses the servo’s built-in position mode with zero speed and acceleration for precise control, ensuring smooth transitions without oscillations.

9.1.6 System Implementation and Details

Safety is ensured through conditional motor unloading and error management. When feedback faults occur—such as communication dropouts—the system illuminates an error LED and stops execution. At startup, a homing routine moves the servos to their neutral positions, enabling the encoders to establish reference baselines.

While running, the lock/release mechanism functions as a safety measure: it relieves cable tension whenever the gripper is open, preventing unintended forces on tissue.

Visual and diagnostic feedback is delivered through a WS2812 LED strip and serial output. Error conditions activate red LEDs, whereas normal operation leaves the indicators off. Both USB and Bluetooth connectivity support data logging, facilitating offline analysis of control performance and user interactions.

The firmware communicates with servos via `HardwareSerial` at 500k baud and uses standard `Serial` for debugging. Motor commands are assembled using a custom protocol that includes unload/enable directives. The main loop executes roughly every 3 ms, striking a balance between system responsiveness and computational overhead.

This implementation embodies a practical tendon-driven control framework, translating high-level intentions into low-level actuation while emphasizing safety and usability within a compact, real-time system.

Chapter 10

Future Works

The future directions of this thesis follow the same two-embodiment structure as the main body. One line of work extends the tendon-driven surgical instrument toward higher precision and richer modularity under clinical constraints. The other extends Lasso Gripper toward broader manipulation scenarios and more capable loop-based grasping strategies. In both cases, the underlying aim is the same: to deepen the mechanism framework developed in this thesis and expand the range of tasks that tendon-driven systems can perform reliably.

10.1 Surgical Instruments

While the current prototype of the cable-driven surgical instrument demonstrates robust functionality, there are significant opportunities for enhancement in precision, adaptability, and overall system intelligence. Future work will focus on the following two key areas:

10.1.1 Enhanced Precision through Adaptive Learning-based Control

A fundamental limitation of the current kinematic model is its assumption of fixed cable lengths. In practice, the stainless steel cables undergo microscopic plastic deformation and experience minute stretching after repeated loading cycles. Even a tiny elongation on the order of 0.1% can accumulate over the cable path, leading to a noticeable positional error at the end-effector, a phenomenon that compromises absolute accuracy and can manifest as a failure to reach a commanded pose.

To address this inherent model inaccuracy, a promising solution is to transition from a purely model-based control scheme to a data-driven, adaptive approach. We propose the integration of Reinforcement Learning (RL) algorithms. An RL agent can be trained, either in a simulated environment or through extensive real-world operation, to learn the subtle, non-linear relationship between motor commands and the resulting end-effector pose, effectively calibrating out the effects of cable stretch and hysteresis in real-time. By continuously correlating the commanded inputs with the actual achieved positions

(potentially validated by external tracking systems during training), the model can self-correct and maintain high precision throughout the instrument’s lifespan, moving beyond the limitations of a static kinematic model.

10.1.2 Expanded Modularity and Context-Aware Operation

The existing quick-swap mechanism provides an excellent foundation for hardware modularity. The next logical step is to exploit this capability by developing a diverse library of specialized end-effector modules (e.g., bipolar forceps, needle holders, scissors, and custom jaws for specific procedures). This would transform the instrument into a versatile multi-tool platform.

To fully realize the potential of this modularity, the system must possess the intelligence to automatically recognize an attached module and reconfigure its control logic accordingly. This can be achieved by embedding a Radio-Frequency Identification (RFID) tag within each module. Upon attachment, a reader in the main handle would instantly identify the module and load the corresponding kinematic profile and software limits (e.g., maximum safe grasping force for delicate tissues). For even greater flexibility and verification, a miniaturized computer vision module could be integrated near the distal end. This would allow the system not only to identify the tool but also to visually confirm its state (e.g., blade open/closed) and relative position to the surgical field, enabling advanced features like semi-autonomous tasks and enhanced safety monitoring.

Together, these advancements in intelligent control and context-aware modularity will pave the way for a new generation of surgical robots that are more precise, adaptable, and intuitive to use.

10.2 Further Applications of Lasso Gripper

Whereas the previous section extends tendon-driven mechanisms toward greater clinical precision and modularity, this section returns to the adaptive grasping side of the thesis. The goal here is not simply to add new demonstrations, but to probe how the loop-based mechanism can be scaled, coordinated, and deployed in more dynamic environments.

Future investigations may involve utilizing multiple loops (two or three) to grasp the same object, potentially enhancing gripping stability and robustness. This multi-loop approach could provide additional security and adaptability in handling complex shapes and sizes, offering a promising avenue for further development.

10.2.1 Multi-Loop Configuration

Implementing a multi-loop configuration involves arranging two or more string loops to work in tandem. Each loop would be controlled independently but coordinated through a central control system to ensure synchronized movement and force

application. This setup is particularly advantageous for gripping objects with irregular geometries or those requiring more secure handling.

10.2.2 Improved Gripping Stability

The use of multiple loops can significantly improve gripping stability. By distributing the gripping force across several contact points, the likelihood of object slippage or deformation is reduced. This is especially beneficial for delicate or fragile items that might be damaged by excessive pressure from a single loop.

10.2.3 Enhanced Adaptability

Multiple loops offer enhanced adaptability in handling a wide range of object sizes and shapes. Each loop can adjust its position and tension dynamically, conforming to the object's contours. This flexibility allows Lasso Gripper to manage complex tasks that single-loop systems might struggle with, such as picking up objects with protrusions or indentations.

10.2.4 Mobile Platform Grasping

As demonstrated in Chapter 6, Lasso Gripper successfully captured flying objects due to its rapid retraction capability. Essentially, this mechanism can be extended to applications involving object securing and transportation. With multiple loops, the gripper significantly increases the likelihood of catching and securing airborne objects.

10.2.5 Potential Challenges and Solutions

While the multi-loop approach offers clear advantages, it also introduces several challenges. Coordinating the movement and tension of multiple loops demands sophisticated control algorithms and real-time feedback systems. Additionally, the mechanical design must prevent interference among the loops during operation. Achieving the ultimate goal of capturing randomly flying objects further requires the grasping algorithm to be robust and adaptive to external obstacles such as walls or floors.

To address these challenges, future research should focus on developing advanced control algorithms that integrate sensor data for precise and dynamic adjustments. Moreover, the mechanical design could incorporate features to facilitate maintenance and quick reconfiguration, ensuring both reliability and operational efficiency. These extensions would further strengthen the central claim of this thesis: tendon-driven mechanisms can be systematically adapted to very different manipulation problems, provided that tension control, routing strategy, and task-specific interaction geometry are designed together.

Chapter 11

Conclusions

This chapter summarizes the main contributions of the thesis: a mechanism-level study of tendon-driven systems, the design and qualitative demonstration of Lasso Gripper as a feasibility platform, and the translation of derived design rules into a handheld laparoscopic instrument. The chapter highlights key findings, acknowledges limitations, and outlines directions for future work.

This study presents a systematic investigation into tendon-driven robotic systems, progressing from fundamental manipulation research to specialized surgical applications. The research demonstrates the versatility and adaptability of cable-driven mechanisms, establishing their significant value across both industrial and medical domains.

The work begins with the development of Lasso Gripper, a novel tendon-driven mechanism that addresses limitations of conventional rigid grippers in handling oversized, irregularly shaped, and delicate objects. Through a launch-and-retraction string loop, the system demonstrates how adaptive capture, distributed contact, and explicit tension management can be combined within a single robotic end-effector. The associated design and experiments establish a set of transferable mechanism insights, including the importance of proximal actuation, compliant interaction, and repeatable tension transmission.

Building upon these mechanism principles, the research then translates the tendon-driven framework into a handheld laparoscopic instrument for minimally invasive surgery. This instrument incorporates a compact, modular architecture with a quick-swap end-effector system, achieving multiple degrees of freedom through precise control of four output cables. The integration of intelligent servos, high-resolution magnetic sensors, and advanced filtering algorithms ensures accurate translation of surgical intent while effectively mitigating physiological tremor, representing a significant advancement in surgical robotics.

A comprehensive experimental framework validates both applications, confirming the effectiveness of the surgical instrument in enhancing surgical dexterity and Lasso Gripper's proficiency in grasping both static and dynamic objects. These parallel developments highlight the cross-domain potential of tendon-driven systems.

Looking forward, two key research directions emerge from this work. First, the integration of data-driven approaches such as Reinforcement Learning can address inherent

system nonlinearities, including cable stretch and hysteresis, to achieve higher precision control. Second, the development of intelligent, auto-identifying modules through technologies like RFID will enable greater versatility and context-aware operation, particularly beneficial for surgical applications.

In conclusion, this research validates the transformative potential of tendon-driven mechanisms through both medical and industrial applications. By bridging fundamental robotic principles with practical clinical and manipulation needs, this work establishes a foundation for future innovations in intelligent, adaptive robotic systems that span healthcare and industrial domains.

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